

SDED 2.0: Vocabulary Index

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This is the compact vocabulary layer of SDED 2.0. It restores the names from earlier Approved Vocabulary libraries and adds explicit new name pairs. Domain and bracketed subdomain headings organize period-separated entries, as in the prior reference format.

The recovered prior library contains **696 distinct name/date mappings**: **438** from the latest Revision B and **258** additional mappings from earlier artifacts. The new library contains **85 proposed name pairs**. An earlier wording is retained as an earlier variant rather than silently disappearing.

Format: Descriptive Name ("Historical Name", est. date). Neither ™ nor (TM) is necessary; either remains acceptable. Every entry has a calendar date, period, or documented bound. "By 1872" means attested no later than 1872; "c." marks approximation. D-numbered notes identify the dated event and sources for every new mapping. A concept publication, a term attestation, and a tradition's development period are different claims. Earlier dates remain historical assertions from the recovered sources, with their verification status explicit; they are not automatically certified by being numeric.

The inherited label *Approved Vocabulary* records the earlier version's status. It does not erase the specific semantic concerns identified in the 2.0 paper. Those mappings remain findable, with a short review note after the libraries. Descriptive guides in Part C are marked as such because they orient the reader without replacing a historical or cultural identity.

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A. New vocabulary additions

These are proposed construction or sense-specific names. The historical alias remains available. In music, seventh and extension intervals are measured above the chord root; mode alterations use major-reference degrees. Scale-degree terms are indexed in their degree sense, not every functional sense. Symbols and actual voicings remain separate.

MUSIC — New Vocabulary

[SEVENTH CHORDS]

Major Triad with Major Seventh ("Major Seventh Chord", est. by 1872) [D01]. Major Triad with Minor Seventh ("Dominant Seventh Chord", est. by 1872) [D02]. Minor Triad with Minor Seventh ("Minor Seventh Chord", est. by 1872) [D03]. Minor Triad with Major Seventh ("Minor-Major Seventh Chord", est. by 1872) [D04]. Diminished Triad with Minor Seventh ("Half-Diminished Seventh Chord", est. by 1872) [D05]. Diminished Triad with Diminished Seventh ("Fully Diminished Seventh Chord", est. by 1872) [D06]. Augmented Triad with Minor Seventh ("Augmented Seventh Chord", est. by 1872) [D07]. Augmented Triad with Major Seventh ("Augmented Major Seventh Chord", est. by 1872) [D08].

[EXTENDED AND SUSPENDED CHORDS]

Major Triad with Minor Seventh and Major Ninth ("Dominant Ninth Chord", est. by 1889) [D09]. Major Triad with Added Major Ninth ("Added-Ninth Major Chord", est. by 1982) [D10]. Major Triad with Major Seventh and Major Ninth ("Major Ninth Chord", est. by 1961) [D11]. Minor Triad with Minor Seventh and Major Ninth ("Minor Ninth Chord", est. by 1961) [D12]. Root-Fourth-Fifth Chord ("Suspended-Fourth Chord", est. by 1982) [D13]. Root-Second-Fifth Chord ("Suspended-Second Chord", est. by 2000) [D14].

[SCALE DEGREES]

Tonic-Relative First Degree ("Tonic", est. 1797) [D15]. Tonic-Relative Second Degree ("Supertonic", est. 1806) [D16]. Tonic-Relative Third Degree ("Mediant", est. 1721) [D17]. Tonic-Relative Fourth Degree ("Subdominant", est. 1775) [D18]. Tonic-Relative Fifth Degree ("Dominant", est. 1819) [D19]. Tonic-Relative Sixth Degree ("Submediant", est. c. 1775) [D20]. Seventh Degree a Semitone Below Tonic ("Leading Tone", est. c. 1889) [D21]. Seventh Degree a Whole Tone Below Tonic ("Subtonic", est. by 2012) [D22].

[INVERSIONS AND BASS]

Root-in-Bass Position ("Root Position", est. by 1889) [D23]. Third-in-Bass Inversion ("First Inversion", est. by 1889) [D24]. Fifth-in-Bass Inversion ("Second Inversion", est. by 1889) [D25]. Seventh-in-Bass Inversion ("Third Inversion", est. by 1889) [D26].

[CHROMATIC HARMONY]

Lowered-Second-Degree Major Triad ("Neapolitan Chord", est. by 1889) [D27]. Lowered-Second-Degree Major Triad with Third in Bass ("Neapolitan Sixth", est. by 1889) [D28]. Augmented-Sixth Sonority with Tonic ("Italian Augmented-Sixth Chord", est. by 1889) [D29]. Augmented-Sixth Sonority with Tonic and Second Degree ("French Augmented-Sixth Chord", est. by 1889) [D30]. Augmented-Sixth Sonority with Tonic and Lowered Third Degree ("German Augmented-Sixth Chord", est. by 1889) [D31].

[MODERN DIATONIC MODES]

Major Diatonic Mode ("Ionian Mode", est. by 1961) [D32]. Diatonic Mode with Lowered Third and Seventh ("Dorian Mode", est. by 1961) [D33]. Diatonic Mode with Lowered Second, Third, Sixth, and Seventh ("Phrygian Mode", est. by 1961) [D34]. Diatonic Mode with Raised Fourth ("Lydian Mode", est. by 1961) [D35]. Diatonic Mode with Lowered Seventh ("Mixolydian Mode", est. by 1961) [D36]. Diatonic Mode with Lowered Third, Sixth, and Seventh ("Aeolian Mode", est. by 1961) [D37]. Diatonic Mode with Lowered Second, Third, Fifth, Sixth, and Seventh ("Locrian Mode", est. by 1961) [D38].

[MINOR-SCALE FORMS]

Natural-Minor Pattern with Raised Seventh ("Harmonic Minor Scale", est. by 1914) [D39]. Natural-Minor Pattern with Raised Sixth and Seventh ("Ascending Melodic Minor Scale", est. by 1914) [D40]. Natural-Minor Pattern with Raised Sixth and Seventh in Both Directions ("Jazz Melodic Minor Scale", est. by 1995) [D41].

[METER]

Two Beats per Bar with Two-Part Beat Subdivision ("Simple Duple Meter", est. by 1914) [D42]. Three Beats per Bar with Two-Part Beat Subdivision ("Simple Triple Meter", est. by 1914) [D43]. Four Beats per Bar with Two-Part Beat Subdivision ("Simple Quadruple Meter", est. by 1914) [D44]. Two Beats per Bar with Three-Part Beat Subdivision ("Compound Duple Meter", est. by 1914) [D45]. Three Beats per Bar with Three-Part Beat Subdivision ("Compound Triple Meter", est. by 1914) [D46]. Four Beats per Bar with Three-Part Beat Subdivision ("Compound Quadruple Meter", est. by 1914) [D47].

[VOICING AND GUITAR]

Root-Third-Seventh Voicing ("Shell Voicing", est. by 1989) [D48]. Second-from-Top Voice Lowered One Octave ("Drop-2 Voicing", est. by 1989) [D49].

[PERFORMANCE AND PRODUCTION]

Pitch Modulation ("Vibrato", est. c. 1876) [D51]. Amplitude Modulation ("Tremolo", est. by 1955) [D52]. Rapid Note Repetition ("Tremolo", est. c. 1801) [D53]. Take-Composite Editing ("Comping", est. by 1994) [D54]. Harmonic Accompaniment ("Comping", est. 1949) [D55].

[NOTATION]

G Clef on the Second Staff Line ("Treble Clef", est. 1667) [D56]. F Clef on the Fourth Staff Line ("Bass Clef", est. 1737) [D57]. C Clef on the Third Staff Line ("Alto Clef", est. 1813) [D58]. C Clef on the Fourth Staff Line ("Tenor Clef", est. 1714) [D59]. Fixed-Pitch Do Solfège ("Fixed-Do Solfège", est. by 1914) [D60]. Tonic-Relative Do Solfège ("Movable-Do Solfège", est. by 1914) [D61].

[TUNING AND TRANSFORMATIONS]

Twelve Equal Logarithmic Divisions of the Octave ("Twelve-Tone Equal Temperament", est. 1584) [D62]. Time-Order Reversal ("Retrograde", est. by 1879) [D63]. Pitch-Direction Inversion ("Contrapuntal Inversion", est. by 1914) [D64].

PHILOSOPHY – New Vocabulary**[PRINCIPLES AND POSITIONS]**

Realism about Abstract Objects ("Platonism", est. 1934–1935) [D72].

[LOGIC]

Deductively Truth-Preserving Argument ("Valid Argument", est. by 1946) [D73]. Valid Argument with True Premises ("Sound Argument", est. by 1946) [D74].

[EPISTEMIC WARRANT]

Justification Independent of Experience ("A Priori Justification", est. 1781–1787) [D75]. Justification Dependent on Experience ("A Posteriori Justification", est. 1781–1787) [D76].

ARTS AND HUMANITIES – New Vocabulary**[RHETORIC]**

Successive-Unit Initial Repetition ("Anaphora", est. before 1101 CE) [D77]. Successive-Unit Final Repetition ("Epistrophe", est. c. 1584) [D78].

[LITERATURE]

Verse without Regular Meter ("Free Verse", est. 1813) [D79].

[THEATRE]

Single-Speaker Extended Speech ("Monologue", est. 1549) [D81].

B. Earlier Approved Vocabulary, carried forward

The following restores the complete latest Revision B library, including its descriptive names, historical names, and domain/subdomain placement. Fourteen inherited date records are corrected in 2.0 across Parts B and D, identified by C-numbered notes. Original date strings remain in the correction ledger and machine-readable provenance. Other dates are carried forward with their inherited status explicit.

MATHEMATICS – Approved Vocabulary – Carried Forward

[ANALYSIS]

Frequency-Decomposition Transform ("Fourier Transform", est. 1822). Curvature-Based Geometry ("Riemannian Geometry", est. 1854). Integral-Summation Series ("Bernoulli Series", est. 1713). Topological-Completeness Space ("Hilbert Space", est. 1902). Complete-Normed Space ("Banach Space", est. 1922). Least-Squares Polynomial ("Legendre Polynomials", est. 1782). Probabilist's-Weight Orthogonal Polynomials ("Hermite Polynomials", est. 1864). Discrete-Time Transform ("Z-Transform", est. 1950). Continuous-Probability Transform ("Laplace Transform", est. 1782). Compactness-Convergence Theorem ("Arzelà-Ascoli Theorem", est. 1895).

[FUNCTIONAL ANALYSIS]

Generalized-Function Distribution ("Schwartz Distribution", est. 1950). Compact-Operator Theory ("Fredholm Theory", est. 1903). Self-Adjoint Extension ("von Neumann Extension", est. 1930). Spectral-Decomposition Theorem ("Spectral Theorem", est. 1930). Dual-Space Representation Theorem ("Riesz Representation Theorem", est. 1907). Weak-Star-Topology Compactness Theorem ("Banach-Alaoglu Theorem", est. 1940).

[ALGEBRA]

Field-Extension Theory ("Galois Theory", est. 1831). Invariant-Transformation Group ("Lie Group", est. 1873). Ring Isomorphism Theorem ("Noether Isomorphism Theorem", est. 1921). Curvature-Defined Smooth Geometry ("Riemannian Geometry", est. 1854). Prime-Power-Subgroup Theorems ("Sylow Theorems", est. 1872). Matrix Normal Form ("Jordan Normal Form", est. 1870). Cofactor-Determinant Expansion ("Laplace Expansion", est. 1772). Unique-Factorization Domain ("Dedekind Domain", est. 1871). Fundamental Isomorphism Theorems ("Noether's Isomorphism Theorems", est. 1927).

[LINEAR ALGEBRA]

Generalized-Matrix Inverse ("Moore-Penrose Inverse", est. 1955). Orthogonalization Process ("Gram-Schmidt Process", est. 1883). Dominant-Eigenvalue Iteration ("Power Iteration", est. 1929). Positive-Definite Decomposition ("Cholesky Decomposition", est. 1910).

[GEOMETRY]

Right-Angle Theorem ("Pythagorean Theorem", est. c. 500 BC). Conic-Section Theorem ("Apollonius Theorem", est. c. 200 BC). Cyclic-Quadrilateral Theorem ("Ptolemy Theorem", est. c. 150). Vertex-to-Side Segment Theorem ("Stewart's Theorem", est. 1746). Vertex-Line Concurrency Theorem ("Ceva's Theorem", est. 1678). Parallel-Intercept Theorem ("Thales' Theorem", est. c. 600 BC). Curvature-Torsion Relations ("Frenet-Serret Formulas", est. 1847).

[DIFFERENTIAL GEOMETRY]

Intrinsic-Curvature Theorem ("Gauss-Bonnet Theorem", est. 1848). Parallel-Transport Connection ("Levi-Civita Connection", est. 1917). Geodesic-Deviation Field ("Jacobi Field", est. 1866). Curvature Tensor ("Riemann Curvature Tensor", est. 1854). Sectional-Curvature Theorem ("Schur's Theorem", est. 1886). Isometric-Manifold Embedding Theorem ("Nash Embedding Theorem", est. 1956).

[TOPOLOGY]

One-Sided Strip ("Möbius Strip", est. 1858). Non-Orientable Bottle ("Klein Bottle", est. 1882). Topological Characteristic ("Euler Characteristic", est. 1752). Continuous-Map Fixed-Point Theorem ("Brouwer Fixed-Point Theorem", est. 1911). Compact-Closed Theorem ("Heine-Borel Theorem", est. 1895). Fundamental Group ("Poincaré Fundamental Group", est. 1895). Homology Numbers ("Betti Numbers", est. 1871). Point-Separation-by-Neighborhoods Space ("Hausdorff Space", est. 1914). Topological-Dimension-Invariance Theorem ("Brouwer Invariance of Domain", est. 1912).

[NUMBER THEORY]

Nontrivial-Exponent Insolubility Theorem ("Fermat's Last Theorem", est. 1637). Prime-Counting Asymptotic Theorem ("Prime Number Theorem", est. 1896). Small-Exponent Congruence Theorem ("Fermat's Little Theorem", est. 1640). Simultaneous-Congruence Solution Theorem ("Chinese Remainder Theorem", est. c. 300). Quadratic-Reciprocity Law ("Gauss Reciprocity", est. 1796). Repeated-Division Greatest-Common-Divisor Algorithm ("Euclidean Algorithm", est. c. 300 BC). Rational-Approximation Theorem ("Dirichlet Approximation", est. 1842). Critical-Line Hypothesis ("Riemann Hypothesis", est. 1859). Arithmetic-Progression Primes Theorem ("Dirichlet's Theorem", est. 1837). Sieve-Bound Theorem ("Brun's Theorem", est. 1919).

[CALCULUS]

Indeterminate-Limit Rule ("L'Hôpital's Rule", est. 1696). Power Series ("Taylor Series", est. 1715). Zero-Centered Series ("Maclaurin Series", est. 1742). Fundamental-Integration Theorem ("Newton-Leibniz Theorem", est. 1684). Product-Rule Integration Formula ("Integration by Parts", est. 1715). Chain-Rule Integration ("Chain Rule Integration", est. 1686). Derivative-Mean-Value Theorem ("Lagrange Mean-Value Theorem", est. 1797). Compact-Continuous Extreme-Value Theorem ("Weierstrass Extreme Value Theorem", est. 1860). Implicit-Function Theorem ("Implicit Differentiation", est. 1878). Dominated-Convergence Theorem ("Lebesgue Dominated Convergence", est. 1904).

[MULTIVARIABLE CALCULUS]

Line-Integral Theorem ("Green's Theorem", est. 1828). Volume-Flux Theorem ("Gauss Divergence Theorem", est. 1813). Surface-Circulation Theorem ("Stokes' Theorem", est. 1854). Variable Transformation ("Jacobian Transformation", est. 1841). Constrained-Optimization Method ("Lagrange Multipliers", est. 1788). Level-Set Implicit-Function Theorem ("Implicit Function Theorem", est. 1878).

[PROBABILITY & STATISTICS]

Conditional-Probability Theorem ("Bayes' Theorem", est. 1763). Normal Distribution ("Gaussian Distribution", est. 1809). Convergence-To-Normal Theorem ("Central Limit Theorem", est. 1922). Sample-Mean-Convergence Law ("Law of Large Numbers", est. 1713). Minimum-Variance Bound ("Cramér-Rao Bound", est. 1946). Maximum-Likelihood Estimation ("Fisher's Maximum Likelihood Estimation [MLE]", est. 1922). Optimal-Test Lemma ("Neyman-Pearson Lemma", est. 1933). Rank-Based Test ("Wilcoxon Test", est. 1945). Goodness-of-Fit Test ("Chi-Square Test", est. 1900). Linear-Correlation Measure ("Pearson Correlation", est. 1895). Rank-Correlation Measure ("Spearman Rank", est. 1904). Likelihood-Curvature Information ("Fisher Information", est. 1925).

[STOCHASTIC PROCESSES]

Memoryless-Transition Process ("Markov Chain", est. 1906). Continuous-Random-Walk Process ("Wiener Process", est. 1923). Discrete-Event Process ("Poisson Process", est. 1837). Stochastic-Integration Calculus ("Itô Calculus", est. 1944). Martingale-Convergence Theorem ("Doob Martingale Convergence Theorem", est. 1953). Optional-Stopping Equation ("Wald's Equation", est. 1945). Stationary-Distribution Theorem ("Perron-Frobenius Theorem", est. 1912).

[LOGIC & SET THEORY]

Set-Theory Axioms ("Zermelo-Fraenkel Axioms", est. 1908). Formal-System Incompleteness Theorems ("Gödel's Incompleteness Theorems", est. 1931). Continuum-Cardinality Hypothesis ("Continuum Hypothesis", est. 1878). Choice-Principle Axiom ("Axiom of Choice", est. 1904). Well-Ordering Theorem ("Zermelo's Theorem", est. 1904). Uncountability Diagonal Argument ("Cantor's Diagonal Argument", est. 1891). Two-Valued Algebra ("Boolean Algebra", est. 1854). Syntactic-Semantic Completeness Theorem ("Gödel Completeness Theorem", est. 1929). Compactness-Logic Theorem ("Compactness Theorem", est. 1930).

PHYSICS – Approved Vocabulary – Carried Forward**[CLASSICAL MECHANICS]**

Classical Mechanics ("Newtonian Mechanics", est. 1687). Inertial-Rest Law ("Newton's First Law", est. 1687). Force-Acceleration Law ("Newton's Second Law", est. 1687). Action-Reaction Law ("Newton's Third Law", est. 1687). Inverse-Square Gravitation ("Newton's Law of Gravitation", est. 1687). Spring-Force Law ("Hooke's Law", est. 1660). Energy-Conservation Mechanics ("Lagrangian Mechanics", est. 1788). Stationary-Action Principle ("Hamilton's Principle", est. 1834). Symmetry-Conservation Theorem ("Noether's Theorem", est. 1918) [C01]. Planetary-Motion Laws ("Kepler's Laws", est. 1609). Rotating-Frame Force ("Coriolis Force", est. 1835). Fictitious-Outward Force ("Centrifugal Force", est. 1659).

[FLUID DYNAMICS]

Pressure-Energy Equation ("Bernoulli's Equation", est. 1738). Viscous-Flow Equations ("Navier-Stokes Equations", est. 1845). Buoyant-Force Principle ("Archimedes' Principle", est. c. 250 BC). Pressure-Transmission Law ("Pascal's Law", est. 1647). Flow-Regime Number ("Reynolds Number", est. 1883). Boundary Layer ("Prandtl Layer", est. 1904). Vorticity-Transport Equation ("Helmholtz Vorticity", est. 1858). Drag-Coefficient Law ("Stokes Drag Law", est. 1851).

[THERMODYNAMICS]

Absolute-Temperature Scale ("Kelvin", est. 1848). Heat-Engine Cycle ("Carnot Cycle", est. 1824). Enthalpy-Additivity Law ("Hess's Law", est. 1840). Throttling-Expansion Effect ("Joule-Thomson Effect", est. 1852). Free Energy ("Gibbs Free Energy", est. 1875). Free Energy ("Helmholtz Free Energy", est. 1882). Real-Gas Equation ("Van der Waals Equation", est. 1873). Magnetic-Transition Temperature ("Curie Temperature", est. 1895). Thermal-Radiation Law ("Stefan-Boltzmann Law", est. 1879). Blackbody-Spectrum Law ("Planck's Law", est. 1900). Peak-Wavelength-Displacement Law ("Wien's Displacement Law", est. 1893).

[STATISTICAL MECHANICS]

Energy Distribution ("Boltzmann Distribution", est. 1868). Velocity Distribution ("Maxwell-Boltzmann Distribution", est. 1860). Fermion Statistics ("Fermi-Dirac Statistics", est. 1926). Boson Statistics ("Bose-Einstein Statistics", est. 1924). Entropy-Microstate Relation ("Boltzmann Entropy", est. 1877). Energy-Equipartition Theorem ("Equipartition Theorem", est. 1859). Canonical Ensemble ("Gibbs Ensemble", est. 1902).

[ELECTROMAGNETISM]

Field-Evolution Equations ("Maxwell's Equations", est. 1865). Electric-Current Unit ("Ampere", est. 1893–1894) [C02]. Electric-Potential Unit ("Volt", est. 1893–1894) [C03]. Magnetic-Flux Unit ("Weber", est. 1930–1935) [C04]. Magnetic-Density Unit ("Tesla", est. 1960) [C05]. Resistance Unit ("Ohm", est. 1893–1894) [C06]. Capacitance Unit ("Farad", est. 1893–1894) [C07]. Inductance Unit ("Henry", est. 1893–1894) [C08]. Current-Voltage Law ("Ohm's Law", est. 1827). Electromagnetic-Induction Law ("Faraday's Law", est. 1831). Magnetic Force ("Lorentz Force", est. 1895). Electrostatic-Force Law ("Coulomb's Law", est. 1785). Magnetic-Field Law ("Ampère's Law", est. 1826). Electric-Flux Law ("Gauss's Law", est. 1835). Electromagnetic-Energy Vector ("Poynting Vector", est. 1884).

[OPTICS]

Refraction-Angle Law ("Snell's Law", est. 1621). Least-Time Principle ("Fermat's Principle", est. 1662). Thin-Lens Equation ("Lensmaker's Equation", est. 1693). Double-Slit Interference ("Young's Interference", est. 1801). Diffraction-Limit Criterion ("Rayleigh Criterion", est. 1879). Polarization-Angle Law ("Brewster's Law", est. 1815). Light Scattering ("Rayleigh Scattering", est. 1871). Frequency-Shift Effect ("Doppler Effect", est. 1842). Fresnel Zones ("Fresnel Zones", est. 1818). Double Refraction ("Double Refraction", est. 1669).

[QUANTUM MECHANICS]

Quantum-Action Constant ("Planck's Constant", est. 1900). Position-Momentum Uncertainty ("Heisenberg Uncertainty", est. 1927). Wavefunction-Evolution Equation ("Schrödinger Equation", est. 1926). Exclusion Principle ("Pauli Exclusion Principle", est. 1925). Quantized-Orbital Model ("Bohr Model", est. 1913). Orbital Quantization ("Bohr-Sommerfeld Quantization", est. 1916). Matter-Wave Relation ("de Broglie Wavelength", est. 1924). Probabilistic-Interpretation Rule ("Born Rule", est. 1926). Matrix-Formulation Mechanics ("Heisenberg Matrix Mechanics", est. 1925). Time-Independent Perturbation Theory ("Rayleigh-Schrödinger Perturbation Theory", est. 1926). Magnetic-Field Splitting ("Zeeman Effect", est. 1896). Quantum-Electrodynamic Shift ("Lamb Shift", est. 1947). Spin-Orbit Coupling ("Fine Structure", est. 1916). Hyperfine-Structure Splitting ("Hyperfine Splitting", est. 1924).

[QUANTUM FIELD THEORY]

Relativistic-Spin Equation ("Dirac Equation", est. 1928). Vacuum-Energy Effect ("Casimir Effect", est. 1948). Mass-Generation Mechanism ("Higgs Mechanism", est. 1964). Electroweak-Unification Model ("Weinberg-Salam Theory", est. 1967). Gauge-Symmetry Theory ("Yang-Mills Theory", est. 1954). Renormalization-Group Flow ("Callan-Symanzik Equation", est. 1970).

[RELATIVITY]

Frame-Transformation Formula ("Lorentz Transformation", est. 1904). Mass-Energy Equivalence (" $E=mc^2$ ", est. 1905). Time-Dilation Effect ("Einstein Time Dilation", est. 1905). Length-Contraction Effect ("Lorentz Contraction", est. 1889). Velocity-Addition Formula ("Einstein Velocity Addition", est. 1905). Spacetime-Curvature Equations ("Einstein Field Equations", est. 1915). Spherical-Solution Metric ("Schwarzschild Metric", est. 1916). Event-Horizon Radius ("Schwarzschild Radius", est. 1916). Cosmological-Expansion Model ("Friedmann Equations", est. 1922). Equivalence-Principle Statement ("Einstein Equivalence", est. 1907). Rotating-Black-Hole Solution ("Kerr Metric", est. 1963).

[ASTROPHYSICS & COSMOLOGY]

Cosmic-Expansion Law ("Hubble's Law", est. 1929). White-Dwarf Mass-Limit ("Chandrasekhar Limit", est. 1931). Neutron-Star Mass-Limit ("Tolman-Oppenheimer-Volkoff Limit", est. 1939). Stellar-Classification Diagram ("Hertzsprung-Russell Diagram", est. 1911). Main-Sequence Duration ("Schönberg-Chandrasekhar Limit", est. 1942). Standard-Candle Supernova ("Type Ia Supernova", est. 1941). Dark-Energy Constant ("Cosmological Constant", est. 1917). Microwave-Background Discovery ("Penzias-Wilson CMB", est. 1965). Rotation-Curve Anomaly ("Dark Matter Problem", est. 1933).

[PARTICLE PHYSICS]

Neutron-Discovery Particle ("Chadwick Neutron", est. 1932). Positron-Discovery Antiparticle ("Anderson Positron", est. 1932). Muon-Discovery Lepton ("Anderson Muon", est. 1936). Parity-Violation Experiment ("Wu Experiment", est. 1956). Quark Model ("Gell-Mann Quark Model", est. 1964). CP-Violation Experiment ("Cronin-Fitch Experiment", est. 1964). Weak-Force Bosons ("W and Z Bosons", est. 1983). Higgs-Boson Discovery ("ATLAS-CMS Discovery", est. 2012). Neutrino Oscillations ("Neutrino Oscillations", est. 1998). CKM-Mixing Matrix ("Cabibbo-Kobayashi-Maskawa Matrix", est. 1973).

[CONDENSED MATTER]

Crystal-Diffraction Law ("Bragg's Law", est. 1913). Zero-Resistance State ("Superconductivity", est. 1911). Josephson-Tunneling Effect ("Josephson Effect", est. 1962). Flux Quantum ("Flux Quantum", est. 1961). Integer-Hall Effect ("Quantum Hall Effect", est. 1980). Fractional-Hall Wavefunction ("Laughlin Wavefunction", est. 1983). BCS-Pairing Theory ("BCS Theory", est. 1957). Crystal-Momentum Theorem ("Bloch Theorem", est. 1928). Interacting-Fermion Quasiparticle Liquid ("Landau Fermi Liquid", est. 1956). Spin Glass ("Spin Glass", est. 1972). Topological Insulators ("Topological Insulators", est. 2007).

COMPUTER SCIENCE – Approved Vocabulary – Carried Forward**[GRAPH ALGORITHMS]**

Shortest-Path Algorithm ("Dijkstra's Algorithm", est. 1956). Shortest-Path With-Negative-Weights ("Bellman-Ford Algorithm", est. 1958). All-Pairs Shortest-Path ("Floyd-Warshall Algorithm", est. 1962). Minimum-Spanning Tree ("Kruskal's Algorithm", est. 1956). Minimum-Spanning-Tree Algorithm ("Prim's Algorithm", est. 1957). Maximum-Flow Algorithm ("Ford-Fulkerson Algorithm", est. 1956). Min-Cut Max-Flow ("Max-Flow Min-Cut Theorem", est. 1956).

[SORTING & SEARCHING ALGORITHMS]

Divide-Merge Sort ("Merge Sort", est. 1945). Partition-Based Sort ("Quick Sort", est. 1960). Heap-Based Sort ("Heap Sort", est. 1964). Binary-Search Algorithm ("Binary Search", est. 1946). Hash-Table Structure ("Hash Table", est. 1953). Balanced-Tree Structure ("AVL Tree", est. 1962).

[COMPUTATIONAL COMPLEXITY]

NP-Completeness Theory ("Cook-Levin Theorem", est. 1971). Polynomial-Time Reduction ("Karp Reduction", est. 1972). Space-Hierarchy Theorem ("Savitch's Theorem", est. 1970).

[INFORMATION THEORY]

Information Entropy ("Shannon Entropy", est. 1948). Channel-Capacity Theorem ("Shannon-Hartley Theorem", est. 1948). Source-Coding Theorem ("Shannon's Source Coding", est. 1948). Error-Correcting Code ("Hamming Code", est. 1950). Optimal-Prefix Coding ("Huffman Coding", est. 1952). Sliding-Window Dictionary Compression ("Lempel-Ziv 77 [LZ77]", est. 1977). Arithmetic Coding ("Arithmetic Coding", est. 1976).

[CRYPTOGRAPHY]

Public-Key Cryptosystem ("Rivest-Shamir-Adleman [RSA]", est. 1977). Asymmetric Key-Exchange ("Diffie-Hellman", est. 1976). Binary-Hash-Tree Structure ("Merkle Tree", est. 1979).

[MACHINE LEARNING & AI]

Machine-Intelligence Test ("Turing Test", est. 1950). Perceptron-Learning Algorithm ("Perceptron", est. 1958). Backpropagation Algorithm ("Backpropagation", est. 1986). Random-Forest Ensemble ("Random Forest", est. 2001). K-Means Clustering ("K-Means", est. 1957). Naive-Bayes Classifier ("Naive Bayes", est. 1960). Decision-Tree Learning ("Decision Trees", est. 1986). PageRank-Link Analysis ("PageRank", est. 1996).

[COMPUTER ARCHITECTURE]

Stored-Program Architecture ("von Neumann Architecture", est. 1945). Separate-Memory Architecture ("Harvard Architecture", est. 1944). Pipeline-Hazard Detection ("Hazard Detection", est. 1964). Dynamic-Instruction Scheduling ("Tomasulo Algorithm", est. 1967). Parallel-Speedup Law ("Amdahl's Law", est. 1967).

[PROGRAMMING LANGUAGES & COMPILERS]

Binary-Production Normal Form ("Chomsky Normal Form", est. 1959). Context-Free-Grammar Metalanguage Notation ("Backus-Naur Form [BNF]", est. 1960). Parametric Type-System ("Hindley-Milner", est. 1978). Untyped Lambda Calculus ("Church Lambda Calculus", est. 1932). Fixed-Point Combinator ("Y Combinator", est. 1930).

[DATABASES & DATA STRUCTURES]

Relational-Database Model ("Codd Relational Model", est. 1970). Normal-Form Decomposition ("Boyce-Codd Normal Form", est. 1974).

[OPERATING SYSTEMS]

Banker's-Algorithm Deadlock ("Banker's Algorithm", est. 1965). Semaphore-Synchronization Primitive ("Dijkstra Semaphore", est. 1965). Monitor-Synchronization Construct ("Hoare Monitor", est. 1974).

ENGINEERING – Approved Vocabulary – Carried Forward**[CONTROL SYSTEMS]**

Frequency-Response Analysis ("Bode Plot", est. 1938). Root-Locus Method ("Evans Root Locus", est. 1948). Stability-Criterion Analysis ("Nyquist Stability", est. 1932). Optimal-State-Estimation Filter ("Kalman Filter", est. 1960). Optimal-Control Maximum Principle ("Pontryagin Maximum Principle", est. 1956). Energy-Function Stability ("Lyapunov Stability", est. 1892).

[SIGNAL PROCESSING]

Sampling-Rate Criterion ("Nyquist-Shannon Theorem", est. 1928). Discrete-Fourier Transform ("Discrete Fourier Transform [DFT]", est. 1965). Fast-Fourier Transform ("Fast Fourier Transform [FFT]", est. 1965). Wavelet-Transform Analysis ("Morlet Wavelet", est. 1984). Adaptive-Filter Algorithm ("Wiener Filter", est. 1949). Cepstral-Analysis Method ("Cepstrum", est. 1963). Hilbert-Transform Analysis ("Hilbert Transform", est. 1905).

[ELECTRICAL CIRCUITS]

Circuit-Analysis Theorem ("Kirchhoff's Laws", est. 1845). Network-Equivalence Theorem ("Thévenin's Theorem", est. 1883). Norton-Equivalent Circuit ("Norton's Theorem", est. 1926). Maximum-Power Transfer ("Maximum Power Theorem", est. 1880). Superposition-Principle Analysis ("Superposition Theorem", est. 1845). Mesh-Current Method ("Mesh Analysis", est. 1847). Nodal-Voltage Method ("Nodal Analysis", est. 1847).

[STRUCTURAL MECHANICS]

Elastic-Deformation Law ("Hooke's Law", est. 1678). Structural-Failure Criterion ("Mohr-Coulomb Theory", est. 1773). Beam-Deflection Theory ("Euler-Bernoulli Beam", est. 1750). Column-Buckling Analysis ("Euler Buckling", est. 1757). Stress-Concentration Factor ("Peterson's Factor", est. 1974). Yield-Criterion Theory ("Von Mises Yield", est. 1913). Plasticity-Flow Theory ("Prandtl-Reuss Flow", est. 1924). Fracture-Mechanics Parameter ("Griffith Criterion", est. 1921).

[MATERIALS SCIENCE]

Crystal-Structure Diffraction ("Bragg's Law", est. 1912). Dislocation Vector ("Burgers Vector", est. 1939). Hardness-Testing Scale ("Mohs Hardness", est. 1822). Indentation-Depth Hardness Scale ("Rockwell Hardness Scale", est. 1914). Pyramidal-Indentation Hardness Test ("Vickers Hardness Test", est. 1921). Phase-Transformation Rule ("Gibbs Phase Rule", est. 1876). Hall-Petch Strengthening ("Hall-Petch Relation", est. 1951).

[FLUID MECHANICS]

Viscous-Flow Equation ("Navier-Stokes Equations", est. 1822). Inviscid-Flow Theory ("Euler Equations", est. 1757). Boundary-Layer Theory ("Prandtl Boundary Layer", est. 1904). Pipe-Flow Friction ("Darcy-Weisbach Equation", est. 1845). Reynolds-Number Similarity ("Reynolds Number", est. 1883). Bernoulli-Energy Conservation ("Bernoulli's Principle", est. 1738). Venturi-Effect Flow ("Venturi Effect", est. 1797). Pitot-Tube Measurement ("Pitot Tube", est. 1732).

[THERMODYNAMICS]

Ideal-Gas Law ("Boyle-Charles Law", est. 1662). Carnot-Efficiency Cycle ("Carnot Cycle", est. 1824). Rankine-Power Cycle ("Rankine Cycle", est. 1859). Brayton-Gas Cycle ("Brayton Cycle", est. 1872). Stirling-Engine Cycle ("Stirling Cycle", est. 1816). Otto-Combustion Cycle ("Otto Cycle", est. 1876). Diesel-Compression Cycle ("Diesel Cycle", est. 1892). Joule-Thomson Effect ("Joule-Thomson Expansion", est. 1852).

MEDICINE – Approved Vocabulary – Carried Forward

[NEURODEGENERATIVE CONDITIONS]

Neural-Degenerative Dementia ("Alzheimer's Disease", est. 1906). Dopaminergic-Motor Disorder ("Parkinson's Disease", est. 1817). Motor-Neuron Degeneration ("Lou Gehrig's Disease", est. 1869). Progressive-Chorea Disorder ("Huntington's Disease", est. 1872). Frontotemporal-Dementia Syndrome ("Pick's Disease", est. 1892). Prion-Encephalopathy Condition ("Creutzfeldt-Jakob Disease", est. 1920). Lewy-Body Dementia ("Lewy Body Disease", est. 1912).

[AUTOIMMUNE CONDITIONS]

Autoimmune-Inflammatory Bowel ("Crohn's Disease", est. 1932). Thyroid-Autoimmune Condition ("Graves' Disease", est. 1835). Hypothyroid-Autoimmune Disorder ("Hashimoto's Thyroiditis", est. 1912). Adrenal-Insufficiency Syndrome ("Addison's Disease", est. 1855). Connective-Tissue Disorder ("Sjögren's Syndrome", est. 1933). Vasculitis-Inflammatory Condition ("Kawasaki Disease", est. 1967). Myasthenic-Neuromuscular Disorder ("Myasthenia Gravis", est. 1877).

[HEMATOLOGIC CONDITIONS]

Lymphatic-Malignancy Condition ("Hodgkin Lymphoma", est. 1832). Non-Hodgkin Lymphoma ("Burkitt Lymphoma", est. 1958). Clotting-Factor Deficiency ("Hemophilia", est. 1803). Von-Willebrand Coagulopathy ("Von Willebrand Disease", est. 1926). Sickle-Cell Hemoglobinopathy ("Sickle Cell Disease", est. 1910). Aplastic-Anemia Syndrome ("Fanconi Anemia", est. 1927). Christmas-Factor Deficiency ("Christmas Disease", est. 1952).

[GENETIC SYNDROMES]

Trisomy-21 Syndrome ("Down Syndrome", est. 1866). Turner-Monosomy Syndrome ("Turner Syndrome", est. 1938). Klinefelter-Aneuploidy Condition ("Klinefelter Syndrome", est. 1942). Fragile-X Syndrome ("Martin-Bell Syndrome", est. 1943). Marfan-Connective Disorder ("Marfan Syndrome", est. 1896). Cystic-Fibrosis CFTR-Mutation ("Cystic Fibrosis", est. 1938). Tay-Sachs Lysosomal Storage ("Tay-Sachs Disease", est. 1881).

[CARDIOVASCULAR CONDITIONS]

Tetralogy-Congenital Defect ("Tetralogy of Fallot", est. 1888). Aortic-Stenosis Syndrome ("Wolff-Parkinson-White Syndrome", est. 1930). Long-QT Arrhythmia ("Romano-Ward Syndrome", est. 1963). Brugada-Arrhythmia Syndrome ("Brugada Syndrome", est. 1992). Takayasu-Arteritis Vasculitis ("Takayasu Arteritis", est. 1908). Kawasaki-Vasculitis Disease ("Kawasaki Disease", est. 1967).

[INFECTIOUS DISEASES]

Bubonic-Plague Bacterium ("Yersinia Pestis", est. 1894). Tuberculosis-Mycobacterium ("Koch's Bacillus", est. 1882). Lyme-Disease Spirochete ("Borrelia Burgdorferi", est. 1982). Epstein-Barr Herpesvirus ("Epstein-Barr Virus [EBV]", est. 1964). Legionnaires-Disease Bacterium ("Legionella", est. 1976). Mycobacterial-Skin Disease ("Hansen's Disease", est. 1873). Chagas-Disease Trypanosomiasis ("Chagas Disease", est. 1909).

[ENDOCRINE CONDITIONS]

Hyperthyroid-Metabolic State ("Graves' Disease", est. 1835). Cushing-Hypercortisolism Syndrome ("Cushing's Syndrome", est. 1912). Acromegaly-Growth Excess ("Acromegaly", est. 1886). Diabetes-Insipidus ADH-Deficiency ("Diabetes Insipidus", est. 1794). Conn-Aldosterone Excess ("Conn's Syndrome", est. 1955). Sheehan-Postpartum Hypopituitarism ("Sheehan's Syndrome", est. 1937).

BIOLOGY – Approved Vocabulary – Carried Forward

[CELLULAR BIOLOGY]

Endoplasmic-Reticulum Network ("ER", est. 1945). Secretory-Vesicle Body ("Golgi Body", est. 1898). Citric-Acid Cycle ("Krebs Cycle", est. 1937). Acetyl-CoA Pathway ("Acetyl Coenzyme-A", est. 1951). Sodium-Potassium Pump ("Na-K-ATPase", est. 1957). Immortal Cells ("HeLa Cells", est. 1951). Replication-Senescence Limit ("Hayflick Limit", est. 1961).

[MOLECULAR GENETICS]

DNA-Polymerase Enzyme ("DNA Pol", est. 1956). Sanger-Sequencing Method ("Sanger Method", est. 1977). PCR-Amplification Technique ("Polymerase Chain Reaction", est. 1983). CRISPR-Gene Editing ("CRISPR-Cas9", est. 2012). Western-Blot Immunoassay ("Western Blot", est. 1979). Southern-Blot Hybridization ("Southern Blot", est. 1975). Northern-Blot RNA-Detection ("Northern Blot", est. 1977). Maxam-Gilbert Sequencing ("Maxam-Gilbert Method", est. 1977).

[EVOLUTIONARY BIOLOGY]

Natural-Selection Theory ("Darwin's Theory", est. 1859). Inheritance Laws ("Mendel's Laws", est. 1866). Hardy-Weinberg Equilibrium ("Hardy-Weinberg Principle", est. 1908). Runaway-Sexual Selection ("Fisherian Runaway", est. 1930). Hybrid-Incompatibility Model ("Dobzhansky-Muller Model", est. 1936). Island-Population Model ("Wright's Island Model", est. 1931). Neutral-Evolution Theory ("Neutral Theory", est. 1968).

[ECOLOGY]

Predator-Prey Equations ("Predator-Prey Model", est. 1925). Ecological-Niche Concept ("Hutchinson's Niche", est. 1957). Competitive-Exclusion Law ("Gause's Law", est. 1934). Island-Biogeography Theory ("MacArthur-Wilson Theory", est. 1967). Tolerance Law ("Shelford's Law", est. 1913). Limiting-Factor Law ("Liebig's Law", est. 1840).

[BIOCHEMISTRY]

Enzyme-Kinetics Model ("Michaelis-Menten Equation", est. 1913). Double-Reciprocal Plot ("Lineweaver-Burk", est. 1934). Protein-Dihedral Plot ("Ramachandran Plot", est. 1963). pH-Buffer Equation ("Henderson-Hasselbalch", est. 1908). Cooperativity-Binding Coefficient ("Hill Equation", est. 1910). Aerobic-Glycolysis Effect ("Warburg Effect", est. 1924). Lactate-Recycling Cycle ("Cori Cycle", est. 1929).

LINGUISTICS AND HUMANITIES – Approved Vocabulary – Carried Forward**[LINGUISTIC THEORY]**

Transformational-Generative Linguistics ("Chomskyan Linguistics", est. 1957). Sign-Structure Linguistics ("Saussurean Linguistics", est. 1916). Universal-Grammar Hypothesis ("Universal Grammar", est. 1965). Linguistic-Relativity Hypothesis ("Linguistic Relativity", est. 1940). Conversational-Maxims Pragmatics ("Grice's Maxims", est. 1975). Communication-Functions Model ("Jakobson Model", est. 1960).

[PHILOSOPHY]

Categorical-Imperative Ethics ("Kantian Ethics", est. 1785). Hedonic-Calculus Utilitarianism ("Benthamite Utilitarianism", est. 1789). Substance-Mind-Body Dualism ("Cartesian Dualism", est. 1641). Transcendent-Form Idealism ("Platonic Forms", est. 380 BCE). Categorical-Term Logic ("Aristotelian Logic", est. 350 BCE). Thesis-Antithesis-Synthesis Dialectic ("Hegelian Dialectic", est. 1807). Will-To-Power Perspectivism ("Nietzschean Philosophy", est. 1886).

[ECONOMIC THEORY]

Demand-Side Economics ("Keynesian Economics", est. 1936). Labor-Value Theory ("Marxist Economics", est. 1867). Invisible-Hand Market Economics ("Smithian Economics", est. 1776). Comparative-Advantage Principle ("Ricardian Comparative Advantage", est. 1817). Population-Growth Theory ("Malthusian Trap", est. 1798). Noncooperative-Equilibrium Game-Theory ("Nash Equilibrium", est. 1950). Allocation-Efficiency Criterion ("Pareto Optimality", est. 1906).

[PSYCHOLOGY]

Unconscious-Psychosexual Psychology ("Freudian Psychology", est. 1900). Classical-Conditioning Model ("Pavlovian Conditioning", est. 1897). Radical Behaviorism ("Skinnerian Behaviorism", est. 1938). Cognitive-Development Stages ("Piagetian Theory", est. 1936). Needs-Hierarchy Model ("Maslow's Hierarchy", est. 1943). Psychosocial-Development Stages ("Erikson's Stages", est. 1950). Collective-Unconscious Archetype Psychology ("Jungian Psychology", est. 1912).

C. Descriptive guides with historical names retained

The left-hand expressions below are orienting descriptions. They are not exact replacement aliases. The named argument, system, or cultural tradition remains the preferred identity where that is its role. These guides are separated from the new replacement names so that the familiar parenthetical display cannot imply false equivalence.

MUSIC – Descriptive Guides

[VOICING AND GUITAR]

Open-Chord-Form Fretboard Mapping ("CAGED System", est. 1974–1975) [D50].

PHILOSOPHY – Descriptive Guides

[PROBLEMS AND THOUGHT EXPERIMENTS]

Justified-True-Belief Insufficiency Problem ("Gettier Problem", est. 1963) [D65]. Descriptive-to-Normative Inference Problem ("Is-Ought Problem", est. 1740) [D66]. Descriptive-to-Normative Inference Problem ("Hume's Guillotine", est. by 1964) [D67]. Artifact-Identity Problem under Replacement and Reassembly ("Ship of Theseus", est. 1655) [D68]. Symbol-Manipulation and Understanding Thought Experiment ("Chinese Room", est. 1980) [D69].

[PRINCIPLES AND POSITIONS]

Parsimony Preference under Adequacy Constraints ("Occam's Razor", est. 1852) [D70]. Mind-Body Substance Dualism in Descartes's Account ("Cartesian Dualism", est. 1641) [D71].

ARTS AND HUMANITIES – Descriptive Guides

[VISUAL ART]

Space around and between Focal Forms ("Negative Space", est. by 1979) [D80].

[ARCHITECTURE]

Pointed-Arch and Rib-Vault Architectural Tradition ("Gothic Architecture", est. 1140–1144) [D82].

[MUSIC CULTURE]

Río de la Plata Music-Dance Tradition ("Tango", est. c. 1875–1900) [D83]. Andalusian Cante-Baile-Toque Tradition ("Flamenco", est. early 1800s) [D84].

[INSTRUMENT CLASSIFICATION]

Sound-Production Instrument Classification ("Hornbostel-Sachs Classification", est. 1914) [D85].

D. Additional names from earlier versions

These distinct earlier mappings supplement Part B. They include alternative descriptive names, additional historical aliases, and different recorded dates. Presence here preserves retrieval and lineage; it does not declare competing variants interchangeable in every use.

MATHEMATICS – Earlier Vocabulary Variants

[ANALYSIS]

Complex-Mapping Framework ("Riemann Geometry", est. 1851). Vector-Norm System ("Banach Space", est. 1922). Least-Squares Approximation ("Legendre Polynomials", est. 1782). Orthogonal-Function Expansion ("Hermite Polynomials", est. 1864). Wave-Packet Transform ("Wavelet Transform", est. 1980).

[FUNCTIONAL ANALYSIS]

Generalized-Function Framework ("Schwartz Distribution", est. 1950). Self-Adjoint Extension Principle ("von Neumann Extension", est. 1930).

[ALGEBRA]

Polynomial-Permutation Structure ("Galois Theory", est. 1831). Field-Construction Framework ("Noether Theorem", est. 1921). Quotient-Group Construction ("Sylow Theorems", est. 1872). Matrix-Decomposition Algorithm ("Jordan Normal Form", est. 1870). Determinant-Expansion Formula ("Laplace Expansion", est. 1772). Prime-Ideal Factorization ("Dedekind Domain", est. 1871).

[LINEAR ALGEBRA]

Least-Squares Solution Method ("Moore-Penrose Inverse", est. 1955). Orthogonal-Decomposition Theorem ("Gram-Schmidt Process", est. 1883). Eigenvalue-Iteration Method ("Power Method", est. 1929). Matrix-Factorization Algorithm ("QR Decomposition", est. 1958). Singular-Value Decomposition ("SVD", est. 1870).

[GEOMETRY]

Right-Angle Relation ("Pythagorean Theorem", est. c. 500 BC). Conic-Section Classification ("Apollonius Theorem", est. c. 200 BC). Circle-Power Theorem ("Ptolemy Theorem", est. c. 150). Angle-Bisector Theorem ("Stewart's Theorem", est. 1746). Triangle-Center Concurrency ("Ceva's Theorem", est. 1678). Rectangular-Coordinate System ("Cartesian Coordinates", est. 1637). Polar-Coordinate System ("Polar Coordinates", est. 1691). Curvature-Torsion Framework ("Frenet-Serret Formulas", est. 1847).

[DIFFERENTIAL GEOMETRY]

Mean-Curvature Formula ("Gauss-Bonnet Theorem", est. 1848). Parallel-Transport Framework ("Levi-Civita Connection", est. 1917). Geodesic-Deviation Equation ("Jacobi Field", est. 1866). Intrinsic-Curvature Tensor ("Riemann Curvature Tensor", est. 1854).

[TOPOLOGY]

One-Sided Surface ("Möbius Strip", est. 1858). Non-Orientable Surface ("Klein Bottle", est. 1882). Manifold-Classification Invariant ("Euler Characteristic", est. 1752). Fixed-Point Theorem ("Brouwer Fixed-Point Theorem", est. 1911). Compactness-Characterization Theorem ("Heine-Borel Theorem", est. 1895). Covering-Space Theory ("Galois Connection", est. 1956). Fundamental-Group Invariant ("Poincaré Fundamental Group", est. 1895).

[NUMBER THEORY]

Nontrivial-Exponent Insolubility ("Fermat's Last Theorem", est. 1637). Prime-Counting Asymptotic ("Prime Number Theorem", est. 1896). Modular-Arithmetic Theorem ("Fermat's Little Theorem", est. 1640). Chinese-Remainder Construction ("Chinese Remainder Theorem", est. c. 300). Greatest-Common-Divisor Algorithm ("Euclidean Algorithm", est. c. 300 BC). Diophantine-Approximation Theorem ("Dirichlet Approximation", est. 1842). Prime-Distribution Hypothesis ("Riemann Hypothesis", est. 1859).

[ANALYTIC NUMBER THEORY]

Complex-Analytic Continuation ("Riemann Zeta Function", est. 1859). Arithmetic-Progression Primes ("Dirichlet's Theorem", est. 1837). Sieve-Method Bound ("Brun's Theorem", est. 1919). Modular-Form L-Function ("Hecke L-Function", est. 1936).

[ARITHMETIC]

Recursive-Ratio Sequence ("Fibonacci Sequence", est. 1202). Prime-Distribution Function ("Gaussian Integer Distribution", est. 1801). Golden-Ratio Constant ("Phidias Constant", est. c. 300 BC). Circle-Circumference Ratio ("Archimedes Constant π ", est. c. 250 BC). Natural-Logarithm Base ("Euler's Number", est. 1731). Imaginary-Unit Constant ("i", est. 1777).

[CALCULUS]

Infinitesimal-Quotient Rule ("L'Hôpital's Rule", est. 1696). Power-Series Expansion ("Taylor Series", est. 1715). Alternating-Series Expansion ("Maclaurin Series", est. 1742). Fundamental-Calculus Theorem ("Newton-Leibniz Theorem", est. 1684). Integration-By-Parts Formula ("Integration by Parts", est. 1715). Change-Of-Variables Theorem ("Substitution Rule", est. 1686). Implicit-Differentiation Rule ("Implicit Function Theorem", est. 1878). Mean-Value Theorem ("Lagrange Mean Value", est. 1797). Extreme-Value Theorem ("Weierstrass Extreme Value", est. 1860).

[MULTIVARIABLE CALCULUS]

Gradient-Field Theorem ("Green's Theorem", est. 1828). Surface-Flux Theorem ("Gauss Divergence Theorem", est. 1813). Circulation-Curl Theorem ("Stokes' Theorem", est. 1854). Multivariable-Chain Rule ("Jacobian Transformation", est. 1841). Lagrange-Multiplier Method ("Constrained Optimization", est. 1788).

[PROBABILITY & STATISTICS]

Inverse-Probability Theorem ("Bayes' Theorem", est. 1763). Normal-Distribution Curve ("Gaussian Distribution", est. 1809). Central-Limit Theorem ("Lindeberg-Lévy CLT", est. 1922). Large-Numbers Law ("Bernoulli's Law", est. 1713). Minimum-Variance Estimator ("Cramér-Rao Bound", est. 1946). Maximum-Likelihood Estimation ("Fisher's MLE", est. 1922). Hypothesis-Testing Framework ("Neyman-Pearson Lemma", est. 1933). Rank-Sum Test ("Wilcoxon Test", est. 1945). Goodness-Of-Fit Test ("Chi-Square Test", est. 1900). Correlation-Coefficient Measure ("Pearson Correlation", est. 1895). Non-Parametric Correlation ("Spearman Rank Correlation", est. 1904).

[STOCHASTIC PROCESSES]

Memoryless-Process Framework ("Markov Chain", est. 1906). Continuous-Diffusion Process ("Wiener Process", est. 1923). Jump-Diffusion Model ("Poisson Process", est. 1837). Stochastic-Calculus Framework ("Itô Calculus", est. 1944). Martingale-Convergence Theorem ("Doob's Theorem", est. 1953).

[LOGIC & SET THEORY]

Axiomatic-Set Framework ("Zermelo-Fraenkel Axioms", est. 1908). Incompleteness-Theorem Pair ("Gödel's Incompleteness Theorems", est. 1931). Continuum-Hypothesis Problem ("Cantor's Hypothesis", est. 1878). Choice-Axiom Principle ("Axiom of Choice", est. 1904). Transfinite-Induction Principle ("Cantor's Theorem", est. 1891). Boolean-Algebra Framework ("Boolean Logic", est. 1854).

[EARLIER VOCABULARY]

Right-Angle Triangle Theorem ("Pythagorean Theorem", est. ~520 BC). Nontrivial Exponent Insolubility Theorem ("Fermat's Last Theorem", est. 1637). Inverse Probability Theorem ("Bayes' Theorem", est. 1763). Greatest Common Divisor Algorithm ("Euclidean Algorithm", est. ~300 BC). Recursive Two-Term Sequence ("Fibonacci Sequence", est. 1202). Infinitesimal Quotient Rule ("L'Hôpital's Rule", est. 1696). Power Series Expansion ("Taylor Series", est. 1715). Rectangular Coordinate System ("Cartesian Coordinates", est. 1637).

PHYSICS – Earlier Vocabulary Variants**[CLASSICAL MECHANICS]**

Inertial-Rest Principle ("Newton's First Law", est. 1687). Spring-Force Proportionality ("Hooke's Law", est. 1660). Energy-Conservation Principle ("Lagrangian Mechanics", est. 1788). Least-Action Principle ("Hamilton's Principle", est. 1834). Angular-Momentum Conservation ("Noether's Theorem", est. 1918) [C09]. Central-Force Motion ("Kepler's Laws", est. 1609). Coriolis-Effect Force ("Coriolis Force", est. 1835). Centrifugal-Force Effect ("Centrifugal Force", est. 1659).

[FLUID DYNAMICS]

Boundary-Layer Theory ("Prandtl Layer", est. 1904).

[THERMODYNAMICS]

Thermal-Energy Scale ("Kelvin", est. 1848). Entropy-Maximization Law ("Second Law", est. 1850). Heat-Engine Efficiency ("Carnot Cycle", est. 1824). Enthalpy-Change Law ("Hess's Law", est. 1840). Gas-Expansion Law ("Joule-Thomson Effect", est. 1852). Free-Energy Potential ("Gibbs Free Energy", est. 1875). Work-Function Potential ("Helmholtz Free Energy", est. 1882). Critical-Temperature Point ("Van der Waals Equation", est. 1873). Ferromagnetic-Transition Temperature ("Curie Temperature", est. 1895). Blackbody-Spectrum Distribution ("Planck's Law", est. 1900). Thermal-Energy Scale Unit ("Kelvin", est. 1848). Entropy-Change Law ("Carnot Cycle", est. 1824).

[STATISTICAL MECHANICS]

Canonical-Ensemble Distribution ("Boltzmann Distribution", est. 1868). Velocity-Distribution Function ("Maxwell-Boltzmann Distribution", est. 1860). Quantum-State Statistics ("Fermi-Dirac Statistics", est. 1926). Boson-Statistics Framework ("Bose-Einstein Statistics", est. 1924). Entropy-Microstates Relation ("Boltzmann Entropy", est. 1877). Equipartition-Energy Theorem ("Maxwell Equipartition", est. 1859).

[ELECTROMAGNETISM]

Electromagnetic-Field Equations ("Maxwell's Equations", est. 1865). Magnetic-Flux-Density Unit ("Tesla", est. 1960) [C10]. Voltage-Current Proportionality ("Ohm's Law", est. 1827). Magnetic-Force Law ("Lorentz Force", est. 1895). Magnetic-Field Circulation ("Ampère's Law", est. 1826). Magnetic-Monopole Absence ("Gauss's Law for Magnetism", est. 1835). Electromagnetic Field Equations ("Maxwell Equations", est. 1865). Magnetic-Flux Density Unit ("Tesla", est. 1960) [C13].

[OPTICS]

Thin-Lens Formula ("Lensmaker's Equation", est. 1693). Wave-Interference Pattern ("Young's Double-Slit", est. 1801). Diffraction-Limit Resolution ("Rayleigh Criterion", est. 1879). Light-Scattering Intensity ("Rayleigh Scattering", est. 1871).

[QUANTUM MECHANICS]

Position-Momentum Uncertainty ("Heisenberg Uncertainty Principle", est. 1927). Spin-Statistics Connection ("Pauli Exclusion Principle", est. 1925). Electron-Energy Levels ("Bohr Model", est. 1913). Orbital-Angular-Momentum Quantization ("Bohr-Sommerfeld Quantization", est. 1916). Matter-Wave Hypothesis ("de Broglie Wavelength", est. 1924). Quantum-State Superposition ("Born Rule", est. 1926). Matrix-Mechanics Formulation ("Heisenberg Matrix Mechanics", est. 1925). Perturbation-Theory Framework ("Rayleigh-Schrödinger Perturbation", est. 1926). Spin-Magnetic-Moment Splitting ("Zeeman Effect", est. 1896). Fine-Structure Splitting ("Lamb Shift", est. 1947).

[QUANTUM FIELD THEORY]

Relativistic-Wave Equation ("Dirac Equation", est. 1928). Electromagnetic-Interaction Theory ("Quantum Electrodynamics", est. 1948). Vacuum-Fluctuation Effect ("Casimir Effect", est. 1948). Symmetry-Breaking Mechanism ("Higgs Mechanism", est. 1964). Quark-Confinement Model ("Quantum Chromodynamics", est. 1973). Electroweak-Unification Theory ("Weinberg-Salam Model", est. 1967).

[RELATIVITY]

Frame-Transformation Equations ("Lorentz Transformations", est. 1904). Relativistic-Velocity Addition ("Einstein Velocity Addition", est. 1905). Gravitational-Time-Dilation Effect ("Schwarzschild Metric", est. 1916). Cosmological-Expansion Solution ("Friedmann Equations", est. 1922). Gravitational-Wave Perturbation ("Gravitational Waves", est. 1916).

[ASTROPHYSICS]

Stellar-Evolution Diagram ("Hertzsprung-Russell Diagram", est. 1911). Main-Sequence Lifetime ("Schönberg-Chandrasekhar Limit", est. 1942). Supernova-Brightness Standard ("Type Ia Supernova", est. 1941). Dark-Energy Hypothesis ("Cosmological Constant", est. 1917). Cosmic-Microwave Background ("Penzias-Wilson Discovery", est. 1965).

[PARTICLE PHYSICS]

Parity-Violation Discovery ("Wu Experiment", est. 1956). Quark-Model Classification ("Gell-Mann Quark Model", est. 1964). CP-Violation Effect ("Cronin-Fitch Experiment", est. 1964). Weak-Interaction Mediator ("W and Z Bosons", est. 1983).

[CONDENSED MATTER]

Zero-Resistance Phenomenon ("Superconductivity", est. 1911). Critical-Current Density ("Josephson Effect", est. 1962). Quantized-Flux Unit ("Flux Quantum", est. 1961). Integer-Hall Conductance ("Quantum Hall Effect", est. 1980). Fractional-Hall States ("Laughlin Wavefunction", est. 1983). Electron-Phonon Coupling ("BCS Theory", est. 1957). Band-Structure Theory ("Bloch Theorem", est. 1928). Fermi-Surface Concept ("Fermi Liquid Theory", est. 1956).

[EARLIER VOCABULARY]

Principle of Inertia ("Newton's First Law", est. 1687). Force–Mass–Acceleration Law ("Newton's Second Law", est. 1687). Action–Reaction Law ("Newton's Third Law", est. 1687). Inverse-Square Gravitation Law ("Newton's Law of Universal Gravitation", est. 1687). Spring Force Proportionality Law ("Hooke's Law", est. 1660). Refraction Sine Law ("Snell's Law", est. 1621). Principle of Least Time ("Fermat's Principle", est. 1662). Fluid Pressure Transmission Law ("Pascal's Law", est. 1647). Buoyant Force Principle ("Archimedes' Principle", est. ~250 BC). Voltage–Current Proportionality Law ("Ohm's Law", est. 1827). Electromagnetic Field Equations ("Maxwell's Equations", est. 1865). Relativistic Transformation Equations ("Lorentz Transformations", est. 1904). Quantum of Action Constant ("Planck's Constant", est. 1900). Uncertainty Principle ("Heisenberg's Uncertainty Principle", est. 1927). Cosmic Expansion Law ("Hubble's Law", est. 1929). White Dwarf Mass Limit ("Chandrasekhar Limit", est. 1931). Ferromagnetic Critical Temperature ("Curie Temperature", est. 1895).

[MECHANICS]

Inertial Field Mechanics ("Newtonian Mechanics", est. 1687). Wave-Interference Equation ("Bernoulli Principle", est. 1738).

CHEMISTRY – Earlier Vocabulary Variants**[EARLIER VOCABULARY]**

Inverse Pressure–Volume Law ("Boyle's Law", est. 1662). Proportional Volume–Temperature Law ("Charles's Law", est. 1787). Mole Particle Constant ("Avogadro's Number", est. by 1909) [C11]. Equilibrium Shift Principle ("Le Châtelier's Principle", est. 1884). Heat Summation Law ("Hess's Law", est. 1840). Absorbance Proportionality Law ("Beer–Lambert Law", est. 1852). Temperature-Dependent Rate Equation ("Arrhenius Equation", est. 1889).

COMPUTER SCIENCE – Earlier Vocabulary Variants**[EARLIER VOCABULARY]**

Public-Key Cryptosystem ("Rivest–Shamir–Adleman (RSA)", est. 1977). Machine Intelligence Imitation Test ("Turing Test", est. 1950). Parallel Speedup Limit Law ("Amdahl's Law", est. 1967). Transistor Density Doubling Trend ("Moore's Law", est. 1965). Network Value Square Law ("Metcalf's Law", est. 1980). Organization Mirroring Principle ("Conway's Law", est. 1968). Stored-Program Computer Architecture ("von Neumann Architecture", est. 1945). Single-Source Shortest Path Algorithm ("Dijkstra's Algorithm", est. 1959).

[ALGORITHMS]

Shortest-Path Optimization Algorithm ("Dijkstra's Algorithm", est. 1956). Divide-Merge Sorting Algorithm ("Merge Sort", est. 1945). Greedy Path Selection Algorithm ("A*", est. 1968).

[INFORMATION THEORY]

Information-Entropy Theory ("Shannon Information Theory", est. 1948). Error-Correction Coding Scheme ("Hamming Code", est. 1950). Compression-Efficiency Model ("Huffman Coding", est. 1952).

[CRYPTOGRAPHY]

Public-Key Encryption Protocol ("RSA", est. 1977). Symmetric-Block Cipher Standard ("AES", est. 1998). Hash-Generation Function ("SHA", est. 1993).

ENG – Earlier Vocabulary Variants**[EARLIER VOCABULARY]**

Linear Elasticity Equation ("Hooke's Law", est. 1660). Fluid Flow Energy Equation ("Bernoulli's Equation", est. 1738). Viscous Flow Field Equations ("Navier–Stokes Equations", est. 1845). Inertial–Viscous Ratio (Re) Number ("Reynolds Number", est. 1883). Harmonic Analysis Transform ("Fourier Transform", est. 1822). Design-Failure Adage ("Murphy's Law", est. 1949). Optimal Estimation Algorithm ("Kalman Filter", est. 1960).

MED – Earlier Vocabulary Variants**[EARLIER VOCABULARY]**

Primary Degenerative Dementia ("Alzheimer's Disease", est. 1906). Idiopathic Motor Neuron Disorder ("Parkinson's Disease", est. 1817). Trisomy 21 Syndrome ("Down Syndrome", est. 1866). Malignant Lymphatic Neoplasm ("Hodgkin Lymphoma", est. 1832). Acute Immune Polyneuropathy ("Guillain–Barré Syndrome", est. 1916). Chronic Intestinal Inflammatory Disease ("Crohn's Disease", est. 1932). Neonatal Vitality Score ("Apgar Score", est. 1952). Glomerular Capsule ("Bowman's Capsule", est. 1842). Uterine Tubes ("Fallopian Tubes", est. 1564).

BIO – Earlier Vocabulary Variants**[EARLIER VOCABULARY]**

Laws of Heredity ("Mendel's Laws", est. 1865). Citric Acid Cycle ("Krebs Cycle", est. 1937). Carbon Fixation Cycle ("Calvin Cycle", est. 1950). Adaptive Evolution Theory ("Darwinian Evolution", est. 1859).

HUMANITIES – Earlier Vocabulary Variants**[EARLIER VOCABULARY]**

Linguistic Relativity Hypothesis ("Sapir–Whorf Hypothesis", est. 1929). Principle of Parsimony ("Occam's Razor", est. 1852) [C12]. Unconscious Speech Error ("Freudian Slip", est. 1901). Needs Pyramid ("Maslow's Hierarchy of Needs", est. 1943). 80/20 Distribution Principle ("Pareto Principle", est. 1906). Assumed Stupidity Principle ("Hanlon's Razor", est. 1980). No-Original-Discoverer Law ("Stigler's Law of Eponymy", est. 1980). Imitation Game ("Turing's Imitation Game", est. 1950).

ENGINEERING – Earlier Vocabulary Variants**[ELECTRICAL]**

Sampling-Rate Criterion ("Nyquist Theorem", est. 1928). Oscillation-Frequency Standard ("Hertz", est. 1930–1935) [C14].

MEDICINE – Earlier Vocabulary Variants**[NEUROLOGY]**

Neural-Degenerative Condition ("Alzheimer's Disease", est. 1906). Motor-System Disorder ("Parkinson's Disease", est. 1817).

[IMMUNOLOGY]

Lymphatic-Proliferation Condition ("Hodgkin Lymphoma", est. 1832). Autoimmune Inflammatory Condition ("Crohn's Disease", est. 1932).

[GENETICS]

Trisomy-21 Condition ("Down Syndrome", est. 1866). Cystic-Fibrosis Condition ("Cystic Fibrosis", est. 1938).

LINGUISTICS AND HUMANITIES — Earlier Vocabulary Variants

[PHILOSOPHY OF LANGUAGE]

Generative-Syntax Framework ("Chomskyan Linguistics", est. 1957). Structural-Semantics Model ("Saussurean Linguistics", est. 1916).

[ECONOMICS]

Macroeconomic-Regulation Framework ("Keynesian Economics", est. 1936). Value-Labor Analysis ("Marxist Economics", est. 1867).

E. Mapping notes

Inherited mappings needing semantic review

The following previously established mappings remain in the index, with their issues explicit: **Unique-Factorization Domain / Dedekind Domain** conflates distinct algebraic properties; **Power Series / Taylor Series** is broader than the intended series type; **Topological-Completeness Space / Hilbert Space** omits the inner-product requirement; **Continuous-Map Fixed-Point Theorem / Brouwer Fixed-Point Theorem** omits distinguishing assumptions; **Energy-Function Stability / Lyapunov Stability** risks substituting a proof method for a property; **Thesis-Antithesis-Synthesis Dialectic / Hegelian Dialectic** compresses a disputed historical interpretation. These are carried forward for continuity, not silently reaffirmed as technically equivalent. See the manuscript's historical critique and its semantic-equivalence rule.

Scope of the new mappings

Historical name	Indexed descriptor and scope
Major Seventh Chord	Major Triad with Major Seventh. Seventh measured above the root; conventional symbol maj7.
Dominant Seventh Chord	Major Triad with Minor Seventh. Chord-quality sense; seventh above root; symbol 7; function separate.
Minor Seventh Chord	Minor Triad with Minor Seventh. Seventh above root; conventional symbol m7.
Minor-Major Seventh Chord	Minor Triad with Major Seventh. Seventh above root; conventional symbol m(maj7).
Half-Diminished Seventh Chord	Diminished Triad with Minor Seventh. Seventh above root; conventional symbol m7(b5).
Fully Diminished Seventh Chord	Diminished Triad with Diminished Seventh. Seventh above root; conventional symbol dim7.
Augmented Seventh Chord	Augmented Triad with Minor Seventh. Seventh above root; conventional symbol 7(#5).
Augmented Major Seventh Chord	Augmented Triad with Major Seventh. Seventh above root; conventional symbol maj7(#5).
Dominant Ninth Chord	Major Triad with Minor Seventh and Major Ninth. Chord construction, not a requirement to voice every tone; symbol 9.
Added-Ninth Major Chord	Major Triad with Added Major Ninth. No seventh included by this construction; symbol add9.
Major Ninth Chord	Major Triad with Major Seventh and Major Ninth. Seventh and ninth above root; symbol maj9.
Minor Ninth Chord	Minor Triad with Minor Seventh and Major Ninth. Seventh and ninth above root; symbol m9.
Suspended-Fourth Chord	Root-Fourth-Fifth Chord. Static sus4 construction, with perfect fourth and fifth; does not assert temporal preparation or resolution.
Suspended-Second Chord	Root-Second-Fifth Chord. Static sus2 construction, with major second and perfect fifth.
Tonic	Tonic-Relative First Degree. Scale-degree sense only; tonic as center or function is a separate use.
Supertonic	Tonic-Relative Second Degree. Scale-degree sense; interval depends on mode or alteration.
Mediant	Tonic-Relative Third Degree. Scale-degree sense only.
Subdominant	Tonic-Relative Fourth Degree. Scale-degree sense, not every subdominant function.
Dominant	Tonic-Relative Fifth Degree. Scale-degree sense, not every harmonic function or seventh-chord quality.
Submediant	Tonic-Relative Sixth Degree. Scale-degree sense only.
Leading Tone	Seventh Degree a Semitone Below Tonic. Standard tonal seventh-degree usage; behavior and chromatic leading tones need context.

Historical name	Indexed descriptor and scope
Subtonic	Seventh Degree a Whole Tone Below Tonic. Standard tonal seventh-degree usage.
Root Position	Root-in-Bass Position. Rooted tertian chord analysis.
First Inversion	Third-in-Bass Inversion. Rooted tertian triad or seventh chord.
Second Inversion	Fifth-in-Bass Inversion. Rooted tertian triad or seventh chord.
Third Inversion	Seventh-in-Bass Inversion. Rooted tertian seventh chord.
Neapolitan Chord	Lowered-Second-Degree Major Triad. Standard tonal chord construction; inversion and predominant use separate.
Neapolitan Sixth	Lowered-Second-Degree Major Triad with Third in Bass. Standard tonal first-inversion construction.
Italian Augmented-Sixth Chord	Augmented-Sixth Sonority with Tonic. Major-reference degrees flat 6, 1, sharp 4; spelling and voice leading matter.
French Augmented-Sixth Chord	Augmented-Sixth Sonority with Tonic and Second Degree. Major-reference degrees flat 6, 1, 2, sharp 4.
German Augmented-Sixth Chord	Augmented-Sixth Sonority with Tonic and Lowered Third Degree. Major-reference degrees flat 6, 1, flat 3, sharp 4; not identical in analysis to an enharmonic dominant seventh.
Ionian Mode	Major Diatonic Mode. Modern tonic-oriented diatonic usage, not an ancient Greek mode.
Dorian Mode	Diatonic Mode with Lowered Third and Seventh. All alterations relative to major-mode degrees.
Phrygian Mode	Diatonic Mode with Lowered Second, Third, Sixth, and Seventh. All alterations relative to major-mode degrees.
Lydian Mode	Diatonic Mode with Raised Fourth. Alteration relative to major-mode degrees.
Mixolydian Mode	Diatonic Mode with Lowered Seventh. Alteration relative to major-mode degrees.
Aeolian Mode	Diatonic Mode with Lowered Third, Sixth, and Seventh. All alterations relative to major-mode degrees.
Locrian Mode	Diatonic Mode with Lowered Second, Third, Fifth, Sixth, and Seventh. All alterations relative to major-mode degrees.
Harmonic Minor Scale	Natural-Minor Pattern with Raised Seventh. Pitch-pattern description; does not restrict melodic use.
Ascending Melodic Minor Scale	Natural-Minor Pattern with Raised Sixth and Seventh. Classical ascending pattern; descending practice is separate.
Jazz Melodic Minor Scale	Natural-Minor Pattern with Raised Sixth and Seventh in Both Directions. Common jazz pitch-pattern usage.
Simple Duple Meter	Two Beats per Bar with Two-Part Beat Subdivision. Primary beat and subdivision under the stated metrical analysis.

Historical name	Indexed descriptor and scope
Simple Triple Meter	Three Beats per Bar with Two-Part Beat Subdivision. Primary beat and subdivision under the stated metrical analysis.
Simple Quadruple Meter	Four Beats per Bar with Two-Part Beat Subdivision. Primary beat and subdivision under the stated metrical analysis.
Compound Duple Meter	Two Beats per Bar with Three-Part Beat Subdivision. For example, an ordinary compound-duple reading of 6/8.
Compound Triple Meter	Three Beats per Bar with Three-Part Beat Subdivision. For example, an ordinary compound-triple reading of 9/8.
Compound Quadruple Meter	Four Beats per Bar with Three-Part Beat Subdivision. For example, an ordinary compound-quadruple reading of 12/8.
Shell Voicing	Root-Third-Seventh Voicing. Specified jazz teaching use; other shell contents exist. Omitting fifth loses some chord-quality distinctions.
Drop-2 Voicing	Second-from-Top Voice Lowered One Octave. Operation on a specified close-position arrangement; not scale degree 2.
CAGED System	Open-Chord-Form Fretboard Mapping. Preserve CAGED as the system's name; C, A, G, E, D identify template forms, not always sounding roots.
Vibrato	Pitch Modulation. Acoustic parameter sense; particular instrument techniques and device labels vary.
Tremolo	Amplitude Modulation. Electronic effect sense; not repeated-note tremolo.
Tremolo	Rapid Note Repetition. Specified performance-technique sense; split from amplitude-modulation effect.
Comping	Take-Composite Editing. Studio editing sense, not accompaniment.
Comping	Harmonic Accompaniment. Jazz accompaniment sense, not studio editing.
Treble Clef	G Clef on the Second Staff Line. Staff lines counted upward.
Bass Clef	F Clef on the Fourth Staff Line. Staff lines counted upward.
Alto Clef	C Clef on the Third Staff Line. Staff lines counted upward.
Tenor Clef	C Clef on the Fourth Staff Line. Staff lines counted upward.
Fixed-Do Solfège	Fixed-Pitch Do Solfège. Pitch-reference policy; local accidental-syllable conventions still require specification.
Movable-Do Solfège	Tonic-Relative Do Solfège. Do-based tonic policy; la-based minor is a separate convention.
Twelve-Tone Equal Temperament	Twelve Equal Logarithmic Divisions of the Octave. Equal frequency ratios, not equal frequency differences.
Retrograde	Time-Order Reversal. Musical sequence operation; specify the sequence being reversed.
Contrapuntal Inversion	Pitch-Direction Inversion. Melodic inversion sense, with axis and interval policy specified; not invertible counterpoint or chord inversion.

Historical name	Indexed descriptor and scope
Gettier Problem	Justified-True-Belief Insufficiency Problem. 1963 dates Gettier's paper, not verified coinage of the later problem label.
Is–Ought Problem	Descriptive-to-Normative Inference Problem. Specify the logical or justificatory claim and interpretation of Hume.
Hume's Guillotine	Descriptive-to-Normative Inference Problem. Later metaphorical label; no first-use date asserted.
Ship of Theseus	Artifact-Identity Problem under Replacement and Reassembly. Retain scenario name; identify the variant under discussion.
Chinese Room	Symbol-Manipulation and Understanding Thought Experiment. 1980 dates Searle's paper; descriptor does not assert the argument's conclusion.
Occam's Razor	Parsimony Preference under Adequacy Constraints. Specify what simplicity measures and what is held equal; not a truth guarantee.
Cartesian Dualism	Mind–Body Substance Dualism in Descartes's Account. 1641 refers to Meditations; wider substance dualism is not an exact historical synonym.
Platonism	Realism about Abstract Objects. Modern metaphysical usage only; not an exact replacement for Plato's historical philosophy.
Valid Argument	Deductively Truth-Preserving Argument. Standard deductive validity sense; does not assert that premises are true.
Sound Argument	Valid Argument with True Premises. Argument sense, not metatheoretic soundness of a proof system.
A Priori Justification	Justification Independent of Experience. Standard epistemic distinction; concept acquisition may still involve experience.
A Posteriori Justification	Justification Dependent on Experience. Epistemic basis, not a simple chronological before/after distinction.
Anaphora	Successive-Unit Initial Repetition. Rhetorical repetition, not linguistic reference.
Epistrophe	Successive-Unit Final Repetition. Rhetorical repetition at the ends of successive units.
Free Verse	Verse without Regular Meter. Orientation to the metrical criterion; does not mean verse without form or rhythm.
Negative Space	Space around and between Focal Forms. Instructional gloss; keep figure–ground ambiguity and compositional uses visible.
Monologue	Single-Speaker Extended Speech. Dramatic speech configuration; does not specify audience or dramatic function.
Gothic Architecture	Pointed-Arch and Rib-Vault Architectural Tradition. Selected identifying features only; not an exact substitute for the historical classification.
Tango	Río de la Plata Music-Dance Tradition. Retain tango as the cultural name; the descriptor is broader and incomplete.

Historical name	Indexed descriptor and scope
Flamenco	Andalusian Cante-Baile-Toque Tradition. Retain flamenco; recognize community histories and the essential Gitano/Roma contribution.
Hornbostel-Sachs Classification	Sound-Production Instrument Classification. Retain the system name and specify edition; not all sound-production classifications are this system.

Recovery coverage

Earlier artifact	Extracted vocabulary entries
build_sded_v1_2_RevisionB_full.py	438
expanded_libraries.py	214
build_sded_v1_2_full.py	70
SDED v1.2_ Specification and Approved Vocabulary Set 1.pdf	70
SDED_Protocol_Specification_v1.2.pdf	438
SDED_v1_2_RevisionB_FULL.pdf	44
SDED_v1_2_full.pdf	70

Counts are occurrences within each source; the same pair can occur in several artifacts. Recovery retains exact descriptive names, historical names, and source date strings. Explicit correction records then determine the current displayed dates. The historical builder is parsed as data and never executed during recovery. These files are recovered v1.2 vocabulary artifacts; no claim is made here that unavailable deposited v1.0/v1.1 PDF text was recovered or independently compared in full.

F. Dates and sources

Research checked September 8, 2026. Each of the 85 new mappings has its own note, including distinct senses of tremolo and comping. The original three publication years are checked alongside the 82 previously unresolved dates.

Reading the dates

An exact year identifies the stated event; it does not imply an exact invention date. A range identifies a documented interval or a sequence of events. “By YEAR” is an upper-bounded historical range: a source establishes existence no later than that year, while an earlier origin remains possible. “C.” preserves approximation, and “early 1800s” preserves the source’s broad early-nineteenth-century period. No arbitrary lower endpoint is invented to make a range appear closed.

The 696 inherited mappings preserve source-artifact provenance. Fourteen receive explicit date corrections below, preserving each previous date in prior_established_date. The remaining 682 retain inherited date assertions, identified by date_status; a complete external-source audit of those dates has not been performed. The

numerical completeness check covers all 781 mappings. This revision supplies external-source date notes for all 85 additions and 14 corrected inherited records. Numeric coverage and historical verification are different measures.

Contributors may submit a short proposal before research is complete. Entry into a published vocabulary requires a date or documented range, the event being dated, and supporting evidence. Blank, unresolved, and unknown date values are rejected by the vocabulary build. Numeric completeness is necessary but cannot establish historical accuracy.

Entry-by-entry evidence

D01. Major Seventh Chord – Major Triad with Major Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D02. Dominant Seventh Chord – Major Triad with Minor Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D03. Minor Seventh Chord – Minor Triad with Minor Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D04. Minor-Major Seventh Chord – Minor Triad with Major Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D05. Half-Diminished Seventh Chord – Diminished Triad with Minor Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D06. Fully Diminished Seventh Chord – Diminished Triad with Diminished Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D07. Augmented Seventh Chord – Augmented Triad with Minor Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: [Henry Wylde, Harmony and the Science of Music \(1872\)](#).

D08. Augmented Major Seventh Chord — Augmented Triad with Major Seventh

by 1872 | construction attestation. The specified triad-plus-seventh construction is listed in Wylde, pp. 105–106. This dates the construction, not every modern label. Sources: Henry Wylde, *Harmony and the Science of Music* (1872).

D09. Dominant Ninth Chord — Major Triad with Minor Seventh and Major Ninth

by 1889 | construction attestation. Major ninth above a dominant seventh: §324. The date does not establish modern function-independent terminology. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D10. Added-Ninth Major Chord — Major Triad with Added Major Ninth

by 1982 | construction attestation. The book explicitly spells added ninth as root, third, fifth, and second/ninth, without a seventh. Sources: Ralph Denyer, *The Guitar Handbook* (1982).

D11. Major Ninth Chord — Major Triad with Major Seventh and Major Ninth

by 1961 | construction attestation. The specified ninth-chord structure occurs in the exhaustive table, example 3-21. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D12. Minor Ninth Chord — Minor Triad with Minor Seventh and Major Ninth

by 1961 | construction attestation. The specified ninth-chord structure occurs in the exhaustive table, example 3-21. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D13. Suspended-Fourth Chord — Root-Fourth-Fifth Chord

by 1982 | construction attestation. The book defines suspended fourth by replacing the third with the fourth. Sources: Ralph Denyer, *The Guitar Handbook* (1982).

D14. Suspended-Second Chord — Root-Second-Fifth Chord

by 2000 | construction attestation. Explicit sus2 construction, illustrated as C–D–G. This is a documented bound, not its invention year. Sources: Rikky Rooksby, *How to Write Songs on Guitar* (2000).

D15. Tonic — Tonic-Relative First Degree

1797 | lexicographic first-known-use report. Dictionary-reported first known English use. Music noun; do not use the adjective's 1649 date. Sources: Merriam-Webster: Tonic.

D16. Supertonic — Tonic-Relative Second Degree

1806 | lexicographic first-known-use report. Dictionary-reported first known English use. Scale-degree noun. Sources: Merriam-Webster: Supertonic.

D17. Mediant — Tonic-Relative Third Degree

1721 | lexicographic first-known-use report. Dictionary-reported first known English use. Scale-degree noun. Sources: Merriam-Webster: Mediant.

D18. Subdominant — Tonic-Relative Fourth Degree

1775 | lexicographic first-known-use report. Dictionary-reported first known English use. Scale-degree noun. Sources: Merriam-Webster: Subdominant.

D19. Dominant — Tonic-Relative Fifth Degree

1819 | lexicographic first-known-use report. Dictionary-reported first known English use. Music noun; do not use the adjective's 1532 date. Sources: Merriam-Webster: Dominant.

D20. Submediant — Tonic-Relative Sixth Degree

c. 1775 | lexicographic first-known-use report. Dictionary-reported first known English use. Scale-degree noun. Sources: Merriam-Webster: Submediant.

D21. Leading Tone — Seventh Degree a Semitone Below Tonic

c. 1889 | lexicographic first-known-use report. Dictionary-reported first known English use. Semitone-below-tonic sense. Sources: Merriam-Webster: Leading Tone.

D22. Subtonic — Seventh Degree a Whole Tone Below Tonic

by 2012 | sense attestation. The scale table explicitly labels the minor seventh “subtonic.” The 1817 dictionary date was rejected because its documented sense is leading tone. Sources: Philip Tagg, *Everyday Tonality*, version 1.3 (2012; first edition 2009).

D23. Root Position — Root-in-Bass Position

by 1889 | construction attestation. Root and inversion positions explicitly taught in §§197, 205–215. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D24. First Inversion — Third-in-Bass Inversion

by 1889 | construction attestation. Root and inversion positions explicitly taught in §§197, 205–215. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D25. Second Inversion — Fifth-in-Bass Inversion

by 1889 | construction attestation. Root and inversion positions explicitly taught in §§197, 205–215. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D26. Third Inversion — Seventh-in-Bass Inversion

by 1889 | construction attestation. Root and inversion positions explicitly taught in §§197, 205–215. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D27. Neapolitan Chord — Lowered-Second-Degree Major Triad

by 1889 | construction attestation. Lowered-second major triad and its named first inversion: §§275–276. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D28. Neapolitan Sixth — Lowered-Second-Degree Major Triad with Third in Bass

by 1889 | construction attestation. Lowered-second major triad and its named first inversion: §§275–276. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D29. Italian Augmented-Sixth Chord — Augmented-Sixth Sonority with Tonic

by 1889 | construction attestation. Named augmented-sixth forms and their contents: §§457, 461, 468. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D30. French Augmented-Sixth Chord — Augmented-Sixth Sonority with Tonic and Second Degree

by 1889 | construction attestation. Named augmented-sixth forms and their contents: §§457, 461, 468. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D31. German Augmented-Sixth Chord — Augmented-Sixth Sonority with Tonic and Lowered Third Degree

by 1889 | construction attestation. Named augmented-sixth forms and their contents: §§457, 461, 468. Sources: Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).

D32. Ionian Mode – Major Diatonic Mode

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D33. Dorian Mode – Diatonic Mode with Lowered Third and Seventh

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D34. Phrygian Mode – Diatonic Mode with Lowered Second, Third, Sixth, and Seventh

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D35. Lydian Mode – Diatonic Mode with Raised Fourth

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D36. Mixolydian Mode – Diatonic Mode with Lowered Seventh

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D37. Aeolian Mode – Diatonic Mode with Lowered Third, Sixth, and Seventh

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D38. Locrian Mode – Diatonic Mode with Lowered Second, Third, Fifth, Sixth, and Seventh

by 1961 | sense attestation. Modern tonic-oriented modal pattern is explicit in example 2-1. Ancient or medieval uses of the name are separate. Sources: Vincent Persichetti, *Twentieth-Century Harmony* (1961).

D39. Harmonic Minor Scale – Natural-Minor Pattern with Raised Seventh

by 1914 | construction attestation. Named harmonic and ascending melodic minor forms, §§86–87. Sources: Karl W. Gehrken, *Music Notation and Terminology* (1914).

D40. Ascending Melodic Minor Scale – Natural-Minor Pattern with Raised Sixth and Seventh

by 1914 | construction attestation. Named harmonic and ascending melodic minor forms, §§86–87. Sources: Karl W. Gehrken, *Music Notation and Terminology* (1914).

D41. Jazz Melodic Minor Scale – Natural-Minor Pattern with Raised Sixth and Seventh in Both Directions

by 1995 | system attestation. Levine's jazz melodic-minor treatment supplies a dated teaching source; it does not date the first jazz performance of the scale. Sources: Mark Levine, *The Jazz Theory Book* (1995); Robert Rawlins, review of *The Jazz Theory Book*, *Music Theory Online* 6.1 (2000).

D42. Simple Duple Meter – Two Beats per Bar with Two-Part Beat Subdivision

by 1914 | classification attestation. §99 explicitly records the English six-cell simple/compound classification. Gehrken also presents a competing classification; these are not conflated. Sources: Karl W. Gehrken, *Music Notation and Terminology* (1914).

D43. Simple Triple Meter – Three Beats per Bar with Two-Part Beat Subdivision

by 1914 | classification attestation. §99 explicitly records the English six-cell simple/compound classification. Gehrken's also presents a competing classification; these are not conflated. Sources: Karl W. Gehrken's, *Music Notation and Terminology* (1914).

D44. Simple Quadruple Meter – Four Beats per Bar with Two-Part Beat Subdivision

by 1914 | classification attestation. §99 explicitly records the English six-cell simple/compound classification. Gehrken's also presents a competing classification; these are not conflated. Sources: Karl W. Gehrken's, *Music Notation and Terminology* (1914).

D45. Compound Duple Meter – Two Beats per Bar with Three-Part Beat Subdivision

by 1914 | classification attestation. §99 explicitly records the English six-cell simple/compound classification. Gehrken's also presents a competing classification; these are not conflated. Sources: Karl W. Gehrken's, *Music Notation and Terminology* (1914).

D46. Compound Triple Meter – Three Beats per Bar with Three-Part Beat Subdivision

by 1914 | classification attestation. §99 explicitly records the English six-cell simple/compound classification. Gehrken's also presents a competing classification; these are not conflated. Sources: Karl W. Gehrken's, *Music Notation and Terminology* (1914).

D47. Compound Quadruple Meter – Four Beats per Bar with Three-Part Beat Subdivision

by 1914 | classification attestation. §99 explicitly records the English six-cell simple/compound classification. Gehrken's also presents a competing classification; these are not conflated. Sources: Karl W. Gehrken's, *Music Notation and Terminology* (1914).

D48. Shell Voicing – Root-Third-Seventh Voicing

by 1989 | construction attestation through scholarly source. Dated construction source: Levine's three-note teaching voicings, identified by McClimon as root-third-seventh shells. This does not date first use of "shell." Sources: Michael McClimon, "Transformations in Tonal Jazz," *Music Theory Online* 23.1 (2017).

D49. Drop-2 Voicing – Second-from-Top Voice Lowered One Octave

by 1989 | construction attestation through scholarly source. Levine's block-chord teaching is the dated source identified in the study. Earliest invention is not asserted. Sources: Jasse Varpama, *Interpretation and Harmonization in Piano Arrangements* (2019).

D50. CAGED System – Open-Chord-Form Fretboard Mapping

1974–1975 | diagram-to-publication range. Diagrams dated 1974, published May 1975, document the five-form method. This does not settle priority or date the later CAGED acronym. Sources: Jon Sievert, "Blue Bear Waltzes," *Guitar Player*, May 1975.

D51. Vibrato – Pitch Modulation

c. 1876 | lexicographic first-known-use report. Dictionary-reported first known English use. Pitch-modulation sense. Sources: Merriam-Webster: *Vibrato*.

D52. Tremolo – Amplitude Modulation

by 1955 | dated artifact. A documented June 1955 amplifier provides a firm artifact bound for electronic amplitude tremolo. Pitch-changing "tremolo" patents were excluded. Sources: Retrofret, inspected Fender Tremolux 5E9, serial 0093; Jeff Owens, *Fender: Pitch or Volume? The Difference Between Tremolo and Vibrato*.

D53. Tremolo — Rapid Note Repetition

c. 1801 | lexicographic first-known-use report. Dictionary-reported first known English use. Rapid-reiteration performance sense, not the later electronic effect. Sources: [Merriam-Webster: Tremolo](#).

D54. Comping — Take-Composite Editing

by 1994 | term attestation. Contemporary studio-production interview uses “vocal comping” in connection with retakes and constructing recorded performances. Sources: [Electronic Musician](#), December 1994.

D55. Comping — Harmonic Accompaniment

1949 | lexicographic first-known-use report. Dictionary-reported first known English use. Accompaniment verb, separately dated from other noun and verb senses. Sources: [Merriam-Webster: Comping](#).

D56. Treble Clef — G Clef on the Second Staff Line

1667 | lexicographic first-known-use report. Dictionary-reported first known English use. G-clef staff-position sense. Sources: [Merriam-Webster: Treble Clef](#).

D57. Bass Clef — F Clef on the Fourth Staff Line

1737 | lexicographic first-known-use report. Dictionary-reported first known English use. F-clef staff-position sense. Sources: [Merriam-Webster: Bass Clef](#).

D58. Alto Clef — C Clef on the Third Staff Line

1813 | lexicographic first-known-use report. Dictionary-reported first known English use. C-clef staff-position sense. Sources: [Merriam-Webster: Alto Clef](#).

D59. Tenor Clef — C Clef on the Fourth Staff Line

1714 | lexicographic first-known-use report. Dictionary-reported first known English use. C-clef staff-position sense. Sources: [Merriam-Webster: Tenor Clef](#).

D60. Fixed-Do Solfège — Fixed-Pitch Do Solfège

by 1914 | system attestation. Fixed-pitch syllables and tonic-relative/do-based-minor usage are documented. Local syllable conventions remain separate. Sources: [Karl W. Gehrkens, Music Notation and Terminology \(1914\)](#).

D61. Movable-Do Solfège — Tonic-Relative Do Solfège

by 1914 | system attestation. Fixed-pitch syllables and tonic-relative/do-based-minor usage are documented. Local syllable conventions remain separate. Sources: [Karl W. Gehrkens, Music Notation and Terminology \(1914\)](#).

D62. Twelve-Tone Equal Temperament — Twelve Equal Logarithmic Divisions of the Octave

1584 | documented calculation. Zhu Zaiyu’s twelve-pitch numerical calculation, as analyzed by Kuttner. Numerical construction is distinguished from later mathematical definitions and disputed priority. Sources: [Fritz A. Kuttner, “Prince Chu Tsai-Yü’s Life and Work,” Ethnomusicology 19.2 \(1975\), 163–206](#).

D63. Retrograde — Time-Order Reversal

by 1879 | construction attestation. Contemporary dictionary documents reversing musical order. The operation is older; this is an attestation bound. Sources: [Frederick Ouseley, “Cancrizans,” Grove’s Dictionary of Music and Musicians, vol. 1 \(1879\)](#).

D64. Contrapuntal Inversion – Pitch-Direction Inversion

by 1914 | construction attestation. Thematic inversion reverses ascending and descending motion, §147. Sources: Karl W. Gehrkens, *Music Notation and Terminology* (1914).

D65. Gettier Problem – Justified-True-Belief Insufficiency Problem

1963 | publication. Publication of Gettier's article; later problem-label coinage is not asserted. Sources: Edmund Gettier, "Is Justified True Belief Knowledge?" *Analysis* 23.6 (1963), 121–123.

D66. Is-Ought Problem – Descriptive-to-Normative Inference Problem

1740 | publication. Publication of the is/ought passage in Book III; 1739 dates the earlier volumes. Sources: David Hume, *A Treatise of Human Nature*, Book III (1740); National Library of Ireland, *Treatise of Human Nature* first-edition record.

D67. Hume's Guillotine – Descriptive-to-Normative Inference Problem

by 1964 | term attestation through scholarly source. Black's paper is the documented source attributed for the later metaphorical label, separate from Hume's 1740 passage. Sources: Max Black, "The Gap Between 'Is' and 'Should'," *Philosophical Review* 73.2 (1964), 165–181; "Giving Up Hume's Guillotine," *Australasian Journal of Philosophy* 93.1 (2015).

D68. Ship of Theseus – Artifact-Identity Problem under Replacement and Reassembly

1655 | variant publication. Hobbes's replacement-and-reassembly variant. This does not date the older replacement-only story attributed to Plutarch. Sources: Thomas Hobbes, *De Corpore* (1655), Part II, chapter XI.

D69. Chinese Room – Symbol-Manipulation and Understanding Thought Experiment

1980 | publication. Publication of Searle's thought experiment. Sources: John Searle, "Minds, Brains, and Programs" (1980).

D70. Occam's Razor – Parsimony Preference under Adequacy Constraints

1852 | lexicographic first-known-use report. Dictionary-reported first known English use. English label, not the medieval principle. Sources: Merriam-Webster: *Occam's Razor*.

D71. Cartesian Dualism – Mind-Body Substance Dualism in Descartes's Account

1641 | publication. Publication of the *Meditations*; dates the specified account, not the later English label. Sources: René Descartes, *Meditations on First Philosophy* (1641).

D72. Platonism – Realism about Abstract Objects

1934–1935 | lecture-to-publication range. Modern mathematical Platonism: Bernays's lecture and publication. This dates an instance of abstract-object realism, not Plato's ancient philosophy. Sources: Paul Bernays, "On Platonism in Mathematics" (1935; lecture 1934).

D73. Valid Argument – Deductively Truth-Preserving Argument

by 1946 | sense attestation through scholarly source. Black's 1946 glossary documents the specified deductive validity/soundness criteria. Hitchcock supplies the earlier attestation, correcting a 1953 attribution. Sources: David Hitchcock, "The origin of the technical use of 'sound argument': a postscript," *Informal Logic* 19 (1999), 213–214.

D74. Sound Argument – Valid Argument with True Premises

by 1946 | sense attestation through scholarly source. Black's 1946 glossary documents the specified deductive validity/soundness criteria. Hitchcock supplies the earlier attestation, correcting a 1953 attribution. Sources: David Hitchcock, "The origin of the technical use of 'sound argument': a postscript," *Informal Logic* 19 (1999), 213–214.

D75. A Priori Justification – Justification Independent of Experience

1781–1787 | edition range. The Critique's first/second-edition formulation of experience-independent and empirical knowledge. Earlier Latin uses do not alone date this epistemic sense. Sources: Immanuel Kant, *Critique of Pure Reason* (1781, 1787).

D76. A Posteriori Justification – Justification Dependent on Experience

1781–1787 | edition range. The Critique's first/second-edition formulation of experience-independent and empirical knowledge. Earlier Latin uses do not alone date this epistemic sense. Sources: Immanuel Kant, *Critique of Pure Reason* (1781, 1787).

D77. Anaphora – Successive-Unit Initial Repetition

before 1101 CE | lexicographic first-known-use report. Dictionary-reported first known English use. Rhetorical sense; dictionary gives "before the 12th century." Sources: Merriam-Webster: Anaphora.

D78. Epistrophe – Successive-Unit Final Repetition

c. 1584 | lexicographic first-known-use report. Dictionary-reported first known English use. Rhetorical repetition sense. Sources: Merriam-Webster: Epistrophe.

D79. Free Verse – Verse without Regular Meter

1813 | lexicographic first-known-use report. Dictionary-reported first known English use. Metrical sense. Sources: Merriam-Webster: Free Verse.

D80. Negative Space – Space around and between Focal Forms

by 1979 | term attestation through edition catalogue. The dated first-edition chapter title documents the instructional term. This is not a claim that artists first used surrounding space in 1979. Sources: Betty Edwards, *Drawing on the Right Side of the Brain* (1979).

D81. Monologue – Single-Speaker Extended Speech

1549 | lexicographic first-known-use report. Dictionary-reported first known English use. Dramatic speech sense. Sources: Merriam-Webster: Monologue.

D82. Gothic Architecture – Pointed-Arch and Rib-Vault Architectural Tradition

1140–1144 | architectural development range. Construction-to-dedication range of the Saint-Denis choir as a formative Gothic example. The later label is not dated to the twelfth century. Sources: The Metropolitan Museum of Art, *The Royal Abbey of Saint-Denis in the Time of Abbot Suger* (1981).

D83. Tango – Río de la Plata Music-Dance Tradition

c. 1875–1900 | historical development range. Late-nineteenth-century formation in the Río de la Plata. Approximate period, not an exact birth date or exclusive national claim. Sources: Buenos Aires cultural heritage research, *Temas de Patrimonio Cultural* 25.

D84. Flamenco — Andalusian Cante-Baile-Toque Tradition

early 1800s | historical development range. Nomination's stated emergence period. Preserve the source's broad interval rather than invent precise endpoints. Sources: Spain, Flamenco nomination to UNESCO, file 00363 (2010).

D85. Hornbostel-Sachs Classification — Sound-Production Instrument Classification

1914 | publication. Publication of the original classification; later revisions require their own edition dates. Sources: Erich M. von Hornbostel and Curt Sachs, *Systematik der Musikinstrumente* (1914).

Corrections to inherited dates

These corrections identify the event the current date represents. They do not establish earliest coinage unless the note explicitly says so. Archival PDFs and builders remain unchanged.

C01. Noether's Theorem — Symmetry-Conservation Theorem

1915 → 1918. Publication of the symmetry/conservation paper; the inherited 1915 value is not its publication date. Sources: Emmy Noether, *Invariante Variationsprobleme* (1918).

C02. Ampere — Electric-Current Unit

1820 → 1893–1894. International electrical-unit standardization and subsequent U.S. legislation. This range dates these defined standards, not earliest coinage. Sources: National Academy of Sciences, *The End of the Nineteenth Century*; Congressional Record, 1894: Units of Electrical Measure.

C03. Volt — Electric-Potential Unit

1860 → 1893–1894. International electrical-unit standardization and subsequent U.S. legislation. This range dates these defined standards, not earliest coinage. Sources: National Academy of Sciences, *The End of the Nineteenth Century*; Congressional Record, 1894: Units of Electrical Measure.

C04. Weber — Magnetic-Flux Unit

1852 → 1930–1935. IEC recommendation of the practical magnetic-flux unit; earlier uses of weber can denote different unit magnitudes. Sources: L. Ruppert, *History of the International Electrotechnical Commission* (1956).

C05. Tesla — Magnetic-Density Unit

1880 → 1960. SI adoption of the tesla as the magnetic-flux-density unit, not a date in Tesla's research career. Sources: BIPM, Resolution 12 of the 11th CGPM (1960).

C06. Ohm — Resistance Unit

1827 → 1893–1894. International electrical-unit standardization and subsequent U.S. legislation. This range dates these defined standards, not earliest coinage. Sources: National Academy of Sciences, *The End of the Nineteenth Century*; Congressional Record, 1894: Units of Electrical Measure.

C07. Farad — Capacitance Unit

1873 → 1893–1894. International electrical-unit standardization and subsequent U.S. legislation. This range dates these defined standards, not earliest coinage. Sources: National Academy of Sciences, *The End of the Nineteenth Century*; Congressional Record, 1894: Units of Electrical Measure.

C08. Henry – Inductance Unit

1832 → 1893–1894. International electrical-unit standardization and subsequent U.S. legislation. This range dates these defined standards, not earliest coinage. Sources: National Academy of Sciences, *The End of the Nineteenth Century*; Congressional Record, 1894: Units of Electrical Measure.

C09. Noether's Theorem – Angular-Momentum Conservation

1915 → 1918. Publication of the symmetry/conservation paper; the inherited 1915 value is not its publication date. Sources: Emmy Noether, *Invariante Variationsprobleme* (1918).

C10. Tesla – Magnetic-Flux-Density Unit

1880 → 1960. SI adoption of the tesla as the magnetic-flux-density unit, not a date in Tesla's research career. Sources: BIPM, Resolution 12 of the 11th CGPM (1960).

C11. Avogadro's Number – Mole Particle Constant

1811 → by 1909. Perrin's named and numerically estimated constant; 1811 concerns Avogadro's gas hypothesis. Sources: Jean Perrin, *Mouvement brownien et grandeurs moléculaires* (1909).

C12. Occam's Razor – Principle of Parsimony

1324 → 1852. Dictionary-reported first known English label; the older principle and medieval writings have separate histories. Sources: Merriam-Webster: *Occam's Razor*.

C13. Tesla – Magnetic-Flux Density Unit

1880 → 1960. SI adoption of the tesla as the magnetic-flux-density unit, not a date in Tesla's research career. Sources: BIPM, Resolution 12 of the 11th CGPM (1960).

C14. Hertz – Oscillation-Frequency Standard

1888 → 1930–1935. IEC naming-recommendation period; not the date of Hertz's experiments. Sources: L. Ruppert, *History of the International Electrotechnical Commission* (1956).

Source locators

- **Henry Wylde, *Harmony and the Science of Music* (1872).** Title page; printed pp. 104–106, especially the complete seventh-chord construction list; pp. 123, 128, 146. Facsimile also available from Wikimedia Commons, Internet Archive item cu31924021751635.
- **Ebenezer Prout, *Harmony: Its Theory and Practice* (1889).** Preface dated June 1889; §§197, 205–215, 275–276, 324, 457, 461, 468. Complete OCR: https://archive.org/download/harmonyitstheory00prouiala/harmonyitstheory00prouiala_djvu.txt
- **Ralph Denyer, *The Guitar Handbook* (1982).** Scanned Pan Books edition: publication page says 1982; chord-construction discussion, printed pp. 130–131. Consult the book itself; the hosting page has an unrelated automated summary.
- **Rikky Rooksby, *How to Write Songs on Guitar* (2000).** First British edition, Balafon Books, 2000; digitized pp. 22–23, “Suspended Second Chords.”
- **Vincent Persichetti, *Twentieth-Century Harmony* (1961).** Printed pp. 31–33, example 2-1, modern modes; p. 77, example 3-21, ninth-chord constructions. The scan was inspected visually.
- **Merriam-Webster: Tonic.** “First Known Use” and the associated sense; consulted September 8, 2026.

- **Merriam-Webster: Supertonic.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Mediant.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Subdominant.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Dominant.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Submediant.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Leading Tone.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Vibrato.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Tremolo.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Comping.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Treble Clef.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Bass Clef.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Alto Clef.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Tenor Clef.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Occam's Razor.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Anaphora.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Epistrophe.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Free Verse.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Merriam-Webster: Monologue.** “First Known Use” and the associated sense; consulted September 8, 2026.
- **Philip Tagg, *Everyday Tonality*, version 1.3 (2012; first edition 2009).** Author-hosted full text; version dated April 6, 2012; chromatic scale table and “Subtonic or leading note?” discussion, printed pp. 34–36.
- **Karl W. Gehrkens, *Music Notation and Terminology* (1914).** Publication page; §§86–87 and 99; footnote 6, discussion of do-based minor and Appendix D; §147.
- **Mark Levine, *The Jazz Theory Book* (1995).** Chapter 3, melodic-minor harmony, pp. 55–70. Publication and chapter coverage are also documented in Robert Rawlins’s contemporary scholarly review, *Music Theory Online* 6.1 (2000).
- **Robert Rawlins, review of *The Jazz Theory Book*, *Music Theory Online* 6.1 (2000).** Bibliographic heading and discussion of Chapter 3, melodic minor.
- **Michael McClimon, “Transformations in Tonal Jazz,” *Music Theory Online* 23.1 (2017).** Footnote 13: identifies root–third–seventh shell voicings and cites Mark Levine, *The Jazz Piano Book* (1989), pp. 17–22.
- **Jasse Varpama, *Interpretation and Harmonization in Piano Arrangements* (2019).** Section 6.4.2, Drop 2; cites Mark Levine, *The Jazz Piano Book* (1989), block-chord chapter, pp. 185 onward.
- **Jon Sievert, “Blue Bear Waltzes,” *Guitar Player*, May 1975.** Printed pp. 26, 30, 41; Keith Allen’s five chord forms and connected fretboard diagrams, themselves dated 1974. Scan visually inspected.
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- **Jeff Owens, Fender: Pitch or Volume? The Difference Between Tremolo and Vibrato.** Manufacturer's explanation distinguishing amplifier amplitude variation from mechanical string-pitch variation.
- **Electronic Musician, December 1994.** "The Gender Gap," interview discussion of studio production, printed p. 77: contemporary use of "vocal comping."
- **Fritz A. Kuttner, "Prince Chu Tsai-Yü's Life and Work," Ethnomusicology 19.2 (1975), 163–206.** Original historical research, especially pp. 163–167 and calculation appendix; searchable reproduction at <https://www.scribd.com/document/414069286/Chinese-Equal-Temperament>.
- **Frederick Ouseley, "Cancrizans," Grove's Dictionary of Music and Musicians, vol. 1 (1879).** Vol. 1, p. 298: backwards canons and retrograde-inverse construction; musical examples included.
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- **"Giving Up Hume's Guillotine," Australasian Journal of Philosophy 93.1 (2015).** Scholarly attribution of the label to Black (1964).
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- **René Descartes, Meditations on First Philosophy (1641).** Meditation VI, real distinction of mind and body; publication year shown on author's works page.
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- **David Hitchcock, "The origin of the technical use of 'sound argument': a postscript," Informal Logic 19 (1999), 213–214.** Original historical investigation reproduces Max Black, Critical Thinking (1946), glossary pp. 387–388, and distinguishes Copi's 1953 formulation.
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